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Architecture, Not Language: Diagnosing Small-Lesion Grounding Failure in Text-Conditioned 3D Medical Localization

MPCS 53113 Natural Language Processing — Final Report University of Chicago

Source code: <https://github.com/rajhansini/Diagnosing-Small-Lesion-Grounding-Failure-in-Text-Conditioned-3D-Medical-Localization>

Abstract

Recent 3D medical vision-language models can detect *that* a pathological finding is present but often fail to say *where* when the finding is small, defaulting to oversized, imprecise regions. This has been reported qualitatively but not quantified as a function of lesion size on volumetric data. We build a text-conditioned contrastive localization pipeline — PubMedBERT sentence embeddings aligned with 3D ResNet-10 patch embeddings on BraTS2020 brain MRI — and measure Dice stratified by true lesion volume across three tumor subregions, running twelve controlled experiments and 171 FDR-corrected paired tests.

The failure is real and specific. Localization collapses monotonically with lesion size (5–15× Dice degradation,

large to small), survives a chance-level control, and replicates in 9 of 9 region×bin comparisons across three independent splits. To rule out the deflationary reading that small lesions are simply hard here, we train a conventional supervised U-Net on the identical data, split and metric: it reaches published BraTS Dice (ET 0.758, TC 0.812, WT 0.851) and degrades by only 1.2-1.3× large-to-small. Small lesions are learnable on this data; text-conditioned grounding is what fails on them.

Roughly half the measured collapse was a metric artifact. Decomposing the pipeline shows the unsupervised Otsu threshold over-predicts lesion volume by 6-223× and is *anti*-correlated with true size ($\rho = -0.26$ to -0.48), mechanically inflating the effect. Under an oracle-volume threshold the large/small ratio falls from 15.0/9.2/4.9 to 14.4/5.6/2.4. A threshold-free pointing game localizes what remains: chance-corrected accuracy is roughly uniform at 46-122× chance everywhere, with one stark exception — for small enhancing tumor the peak response lands inside the lesion in **0 of 21 patients**, exactly chance.

The language pathway contributes almost nothing. Replacing PubMedBERT with general-domain BERT, with a randomly initialized never-trained BERT, or with random vectors carrying no language at all produces effects that change sign across training runs — indistinguishable from the 0.0044 Dice noise floor of simply re-training the baseline. Only discarding PubMedBERT’s anisotropic embedding *geometry* reliably hurts (-0.0087 , consistent across three seeds). Probing the trained model directly, negating the query, destroying its word order, swapping its anatomical referent, or replacing it with contentless filler all leave the projected embedding above 0.94 cosine to the original and the heatmap at $\rho = 0.56-1.00$; for whole tumor, generic filler and a wrong-region term both *outperform* the true clinical description. The query functions as an opaque class identifier that happens to be spelled in English.

No intervention fixed the failure, and the one that appeared to was a threshold artifact. Size-conditioned prompting, multi-scale ensembling, scale-matched retraining (which a noise probe shows learned resize-interpolation shortcuts, ANOVA $p \approx 4 \times 10^{-251}$) and a uniformity regularizer all fail to beat the plain baseline. Evaluating the frozen model at a smaller query window appeared to win decisively — until re-scored under better binarizers, where its +0.044 Dice advantage inverts to -0.039 (deployable top-1%) and -0.043 (oracle). It does, however, genuinely improve *pointing* accuracy for enhancing tumor in every size bin, lifting the previously-at-chance small-ET bin to 309× chance.

The durable contribution is therefore methodological: a set of controls — chance baselines, noise probes, cross-seed replication at the level of the training run, pipeline decomposition, metric triangulation and metric well-definedness — that between them overturned four conclusions this work had already written down as findings.

1. Introduction

Radiology reports routinely describe findings in natural language — location, size, character — while the underlying evidence lives in a 3D volume (CT/MRI). A model that could ground free text directly onto the matching 3D region would be useful for report-to-image verification, weakly-supervised segmentation without costly voxel-level labels, and explainable AI-assisted diagnosis. Several recent 3D medical vision-language models attempt exactly this, and a consistent, troubling pattern has emerged in the literature: these models can often tell *that* a finding exists, but when the finding is small, they fail to say *where* — collapsing to an imprecise region spanning a whole organ or quadrant rather than the actual lesion.

This is not a cosmetic failure. Small, subtle findings are disproportionately the clinically important case: large, obvious masses rarely need AI assistance to be found, while small or early-stage lesions are exactly where a grounding tool would be most valuable — and exactly where current systems are documented to fail worst. If this failure mode is real but unquantified, it’s difficult to know how bad it is, whether it’s fixable, or what a fix would even target.

This project does three things. First, it builds a controlled experimental setup to **rigorously quantify** this failure as a function of lesion size, rather than relying on qualitative or anecdotal reports — measuring localization quality (Dice/IoU) stratified into small/medium/large tercile bins, on a held-out validation set, across three independently-defined tumor subregions in the BraTS2020 dataset. Second, it runs a systematic **series of ablation studies** — text-side, inference-side, and training-side interventions — to see whether the failure is a *language* problem, an *architecture* problem, or something that can be trained away. Third, throughout, it

applies a consistent set of **diagnostic tools** (chance-level controls, shortcut-learning noise probes, cross-split replication, family-wise multiple-comparisons correction) so that every claim of “improvement” or “failure” is verified rather than eyeballed. As detailed below, the core failure turned out to be architectural rather than linguistic, and no single ablation fully solves it — which is itself a useful, honest finding for anyone building on this line of work.

2. Problem Definition

Given a 3D medical volume V and a text description t of a finding, a text-conditioned localization model $f(V, t) \rightarrow M$ predicts a spatial region M corresponding to that finding. Let M^* be the ground-truth region with voxel volume $|M^*|$. We study the relationship between localization quality ($\text{Dice}(M, M^*)$) and $|M^*|$: does quality degrade smoothly, or collapse below some size threshold? This has been noted as a real limitation in recent 3D medical VLM literature but not rigorously measured in a controlled, size-stratified way on volumetric data — that is the gap this project addresses.

3. Related Work

Contrastive vision-language pretraining. CLIP (Radford et al., 2021) established the pattern this project builds on: align image and text embeddings in a shared space via contrastive learning, then use the aligned space for zero-shot classification, retrieval, or (with post-hoc techniques) localization. We adapt this pattern to 3D volumetric patches rather than 2D natural images.

BERT sentence embeddings and anisotropy. Reimers & Gurevych’s Sentence-BERT (2019) showed that raw BERT embeddings (CLS-token or mean-pooled) cluster tightly in a narrow cone of the embedding space and perform poorly on semantic similarity tasks without task-specific fine-tuning — a property called anisotropy. We observed exactly this with PubMedBERT (Section 5): our four class descriptions have ~ 0.99 pairwise cosine similarity in the raw embedding space, confirming the phenomenon holds for biomedical BERT variants too, not just general-domain BERT.

Text-driven medical segmentation. MedCLIP-SAMv2 and SimTxtSeg are recent frameworks that use text prompts to drive weakly-supervised or zero-shot medical image segmentation, integrating CLIP-style models with SAM-style segmentation. These operate primarily on 2D slices or 2D imaging modalities (e.g. chest X-ray, 2D CT slices); this project instead works with genuinely volumetric 3D patches and a sliding-window localization mechanism suited to that setting.

3D medical vision-language grounding failures. Several 2025-2026 papers document the specific failure this project quantifies. A benchmark of 3D medical VQA models found that “without explicit spatial localization, VLMs fail to attend to subtle lesion signals in raw 3D volumes... for small targets such as lesions or nodules, models default to large bounding boxes encompassing the entire organ or image quadrant.” An audit of frontier medical VLMs similarly found grounding to be “a major failure point,” with small-lesion measurement specifically flagged as harder than existence detection due to limited annotated small-lesion cases. These papers establish that the failure is real and current, but report it qualitatively or as one metric among many in a broader benchmark — none isolate the *relationship between lesion size and localization quality* with a controlled chance-level comparison, which is the specific gap this project fills. That comparison matters because Dice is known to be geometrically harsher on small structures for *any* method (Section 6) — without a chance baseline, a raw “small lesions score worse” result is not yet evidence of a model-specific failure, just a property of the metric. This project’s contribution is showing the failure survives that control.

BraTS. The BraTS challenge (Menze et al., 2015; continued annually since) is the standard benchmark for brain tumor segmentation from multi-modal MRI, and is the data source for this project (2020 edition).

4. Dataset

BraTS2020 (Kaggle: awsaf49/brats20-dataset-training-validation), sourced from the MICCAI Brain Tumor Segmentation Challenge. 369 glioma patients, each with 4 co-registered, skull-stripped MRI sequences (T1,

T1ce, T2, FLAIR) and a voxel-level expert segmentation mask with three standard evaluation regions: - **ET** (enhancing tumor) - **TC** (tumor core = $ET \cup \text{necrotic core}$) - **WT** (whole tumor = $TC \cup \text{peritumoral edema}$)

One patient (BraTS20_Training_355) ships with a non-standard segmentation filename (w39_1998.09.19_Segm.nii), a leftover artifact from the original hospital anonymization process never corrected in this release — handled with a filename fallback in preprocessing (see Challenges).

All volumes were intensity-normalized (z-score within the brain mask, per modality) and resampled to 128^3 . True lesion volumes (mm^3) were computed from the native-resolution segmentation and voxel spacing (not the resampled grid), then used to bin patients into small/medium/large terciles per region:

Region	Small cutoff	Large cutoff	Patients with zero volume
ET	$\leq 9,051 \text{ mm}^3$	$> 26,245 \text{ mm}^3$	27/369 (no enhancing component)
TC	$\leq 19,191 \text{ mm}^3$	$> 49,948 \text{ mm}^3$	0/369
WT	$\leq 63,126 \text{ mm}^3$	$> 125,309 \text{ mm}^3$	0/369

Split: 296 train / 73 held-out validation patients (80/20, fixed seed).

5. Method

Text encoder: microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract-fulltext, mean-pooled over non-padding tokens. Four base region descriptions (ET/TC/WT/NONE, ~3 template sentences each). Raw PubMedBERT sentence embeddings for these four classes have ~0.99 pairwise cosine similarity — BERT anisotropy (Section 3), not a bug. This means the *trainable projection head*, not the frozen text encoder, is responsible for introducing discriminability.

Volume encoder: MONAI 3D ResNet-10, 4 input channels, operating on 32^3 -voxel patches sampled from the 128^3 volume (positive patches centered on a random voxel within the target region’s mask; a background/“NONE” class sampled from outside the whole-tumor region).

Alignment (P’ baseline): contrastive classification — image and text embeddings are projected into a shared 256-d space (L2-normalized), and trained with cross-entropy over cosine-similarity logits (temperature 0.07) against the 4 classes (ET/TC/WT/NONE). This is the reproducibility checkpoint (P’): if the model can’t learn to distinguish these four classes well above chance, nothing downstream is trustworthy.

Localization / heatmap extraction: Grad-CAM was the original plan but was rejected on inspection (Section 8) — this architecture globally average-pools each 32^3 patch to a single embedding, leaving no meaningful spatial feature map near the output to back-propagate onto. Instead, we use a **sliding-window similarity map**: the trained patch encoder is swept across the full 128^3 volume (stride 16), and each window’s cosine similarity to the query text embedding is accumulated into a per-voxel heatmap. Validated with a sanity check confirming the ET-query heatmap scores higher inside the true ET region than outside (+0.247 mean difference) and the inverse holds for the NONE query.

Binarization: Otsu’s method (unsupervised, per-volume) converts the continuous heatmap into a predicted mask for Dice/IoU scoring — chosen so the threshold is not tuned against ground truth (which would leak test-time information).

5.1 Techniques used, and how they work

Several methods this project relies on were not covered in class. Each is summarized here, with a reference to a fuller description.

Contrastive alignment with a temperature-scaled softmax (InfoNCE). Image and text embeddings are L2-normalized and projected into a shared space; the cosine similarity between an image embedding and each of the K class text embeddings forms a logit vector, divided by a temperature τ (here 0.07) and passed through

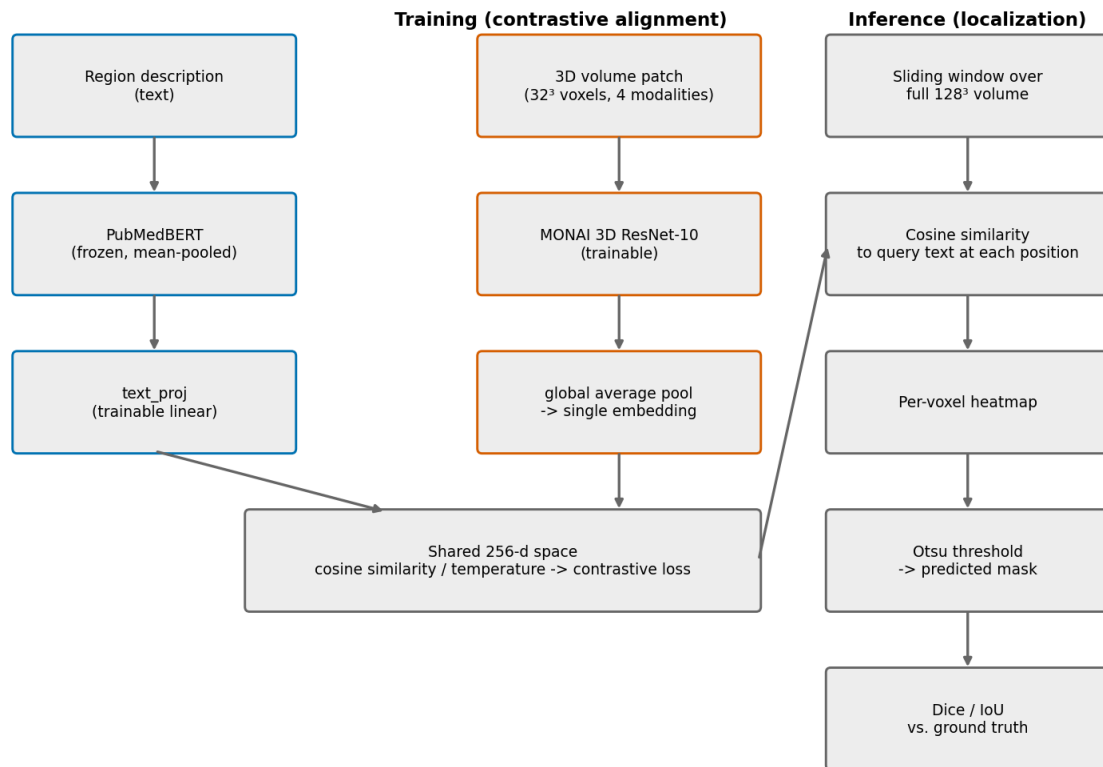


Figure 1: Pipeline schematic: contrastive text-volume alignment at training time (left/center), and sliding-window heatmap extraction at inference time (right).

softmax cross-entropy against the true class:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(v, t_y)/\tau)}{\sum_{k=1}^K \exp(\text{sim}(v, t_k)/\tau)}$$

Low τ sharpens the distribution, penalizing near-misses heavily and pushing classes apart. This is the CLIP/InfoNCE objective (Oord et al., 2018; Radford et al., 2021) with the text side acting as a fixed set of class prototypes rather than as in-batch negatives.

Otsu’s method (Otsu, 1979). An unsupervised way to pick a binarization threshold from a single image’s intensity histogram. It considers every candidate cut point and selects the one maximizing the variance *between* the two resulting groups — equivalently minimizing variance within them. We chose it precisely because it never sees ground truth, so it cannot leak test-time information into Dice. Section 6.3 quantifies what that choice cost.

The uniformity regularizer (Wang & Isola, 2020). Wang and Isola showed contrastive representation quality decomposes into *alignment* (matched pairs are close) and *uniformity* (embeddings spread over the hypersphere rather than collapsing). RQ4 produced exactly a uniformity failure — the class embeddings collapsed into a hub. RQ6 adds a penalty on high pairwise cosine similarity among projected class embeddings to push them apart.

The pointing game (Zhang et al., 2018). A localization metric from weakly-supervised grounding: take the single highest-response location in a saliency map and ask whether it falls inside the ground-truth region. Unlike Dice it is threshold-free and carries no geometric penalty against small targets, so it separates “did the model point at the lesion” from “did it draw the right boundary.” Its chance level still scales with target size, so we always report lift over a per-bin chance baseline rather than the raw hit rate.

Wilcoxon signed-rank test (Wilcoxon, 1945). A non-parametric paired test. Because each patient is scored under both conditions, comparisons are paired; and because per-patient Dice is bounded, skewed, and frequently near zero for small lesions, a paired *t*-test’s normality assumption is not safe. Wilcoxon ranks the absolute paired differences and tests whether positive and negative ranks are balanced.

Benjamini-Hochberg FDR correction (Benjamini & Hochberg, 1995). With 171 accumulated tests at $\alpha=0.05$, roughly nine “significant” results would be expected from noise alone. BH sorts the m p-values ascending and rejects the largest i for which $p_{(i)} \leq \frac{i}{m}\alpha$, controlling the expected *proportion* of false discoveries among rejections. It is less conservative than Bonferroni, which is appropriate here because these tests are positively correlated (they share a baseline arm).

Matched-pairs rank-biserial correlation and bootstrap intervals. Effect size for Wilcoxon, ranging from -1 to $+1$. Reported alongside every p-value because with $n=20-32$ per bin a result can be significant yet negligible; Section 7 flags such cases explicitly. Confidence intervals on mean paired differences use a percentile bootstrap (10,000 resamples, seeded) rather than a normal-theory interval, for the same distributional reason Wilcoxon is used.

5.2 Code components

All code was written for this project; see `src/README.md` for a file-by-file listing with line counts. The major components:

Component	Files	Role
Data pipeline	<code>preprocess.py</code> , <code>text_encoder.py</code>	NIfTI loading, per-modality z-scoring inside the brain mask, resampling to 128^3 , true native-resolution lesion volumes for size binning; PubMedBERT embedding of all text variants.

Component	Files	Role
Datasets	dataset.py, dataset_rq2.py, dataset_rq4.py, dataset_pprime.py	Patch samplers for each experimental condition (region-labeled, size-conditioned, scale-matched) plus the full-volume segmentation loader for P'. region_mask() here is the single definition of ET/TC/WT used everywhere.
Model	model.py	TextVolumeAligner: MONAI 3D ResNet-10 volume encoder plus a linear text projection into a shared 256-d L2-normalized space.
Localization	localize.py	sliding_window_heatmap(): sweeps the encoder across the volume, accumulating per-voxel cosine similarity to a query. Supports querying at a different physical window size than the model was trained at, which is what makes the whole RQ3/RQ3b/RQ3c/RQ12 window sweep possible without retraining.
Training	train_baseline.py, train_rq2/4/5/6/7.py, train_pprime_supervised.py	One script per experimental arm, each taking --seed to control the train/val split for cross-seed replication.
Evaluation	evaluate_rq1.py and 10 siblings	evaluate_rq1.py defines otsu_threshold(), dice_iou() and size_bin(), which every other evaluation — including the supervised P' — imports, so all arms are scored by literally the same code.
Diagnostics	test_rq4_shortcut_hypothesis.py, test_rq6_hub_bias.py, compute_chance_baseline.py, sanity_check_localize.py	Noise probes, chance-level control, and the pre-flight check that the heatmap scores higher inside the true region than outside.
Analysis	analyze_full_family.py, analyze_seed_replication.py, analyze_rq7_multiseed.py, and 5 others	Recompute every statistic directly from the saved per-patient CSVs, so no number in this report is transcribed by hand.
Figures	9 make_figure*.py scripts	Every figure regenerates from the result CSVs.

Two recurring design decisions are worth naming. First, **evaluation scripts import their metric definitions rather than redefining them**, which is what makes cross-experiment comparison meaningful. Second, **several scripts carry a built-in correctness gate**: evaluate_rq8_compositionality.py asserts its “original” condition reproduces RQ1’s CSV exactly, and evaluate_grounding_sweep.py at window 32 must reproduce RQ11’s — both

caught real bugs during development.

6. Core Result: Size-Stratified Localization Failure

Note on multiple comparisons. Sections 6 and 7 together report many paired significance tests (171 in total by the end of Section 7: region $\times$ size-bin $\times$ comparison, across 8 research questions). At uncorrected $\alpha=0.05$, that volume of testing would be expected to produce a small number of spurious “significant” results by chance alone. We therefore apply Benjamini-Hochberg FDR correction across the full accumulated family of tests and report both the raw p-value and BH-adjusted q-value wherever a specific claim rests on statistical significance; any result that is significant raw but does not survive correction is explicitly flagged as such rather than presented as a finding.

6.1 P’: validating the pipeline against a previously-studied problem

Every result in this report is produced by one pipeline — our preprocessing, our split, our `dice_iou()`, our size terciles. If any of those carried a bug, “small lesions localize badly” would be unfalsifiable from the inside: we would have no way to distinguish a real grounding failure from, say, a misaligned mask or a broken metric. Two checks address this, an internal one and an external one.

Internal: does the alignment learn anything? Full run, 296 train / 73 val patients, 30 epochs: validation accuracy on the 4-way ET/TC/WT/NONE classification rose from 0.51 to a peak of 0.671 (epoch 26), finishing at 0.626, against a chance level of 0.25. The pipeline learns a genuine, non-trivial text-volume alignment signal.

External (P’): does the same machinery reproduce published results on a problem someone else has already solved? That problem is supervised BraTS tumor segmentation, scored publicly by the MICCAI challenge since 2012. We trained a plain 3D U-Net (Ronneberger et al., 2015) reusing, unchanged, the same preprocessed volumes, the same seed-0 split, the same `region_mask()` definitions and the same `dice_iou()` — changing only the model and the supervision signal.

Region	P’ Dice (ours)	Published BraTS2020 range	In range?
ET	0.758	0.70 – 0.80	yes
TC	0.812	0.80 – 0.87	yes
WT	0.851	0.86 – 0.89	0.009 below

Two of three regions land inside the published range and the third is 0.009 short of it, which is what a single small U-Net on 128^3 resampled volumes should look like against leaderboard entries that use native resolution, ensembling and test-time augmentation. **The shared machinery is therefore sound, and the low absolute Dice in Sections 6.2-7 is a property of text-conditioned localization rather than a defect in the plumbing underneath it.**

P’ also answers a question no text-conditioned arm could. Because it is scored by the identical size terciles, its large/small ratio is directly comparable to RQ1’s — which separates “text-conditioned grounding fails on small lesions” from “small lesions are simply hard for everything on this data.”

Region	P’ supervised L/S	RQ1 text-conditioned	
		L/S (Otsu)	RQ1 (oracle threshold)
ET	1.3×	15.0×	14.4×
TC	1.3×	9.2×	5.6×
WT	1.2×	4.9×	2.4×

A fully supervised model on this exact data barely degrades at all: $0.636 \rightarrow 0.858$ Dice across ET’s size range, against the text-conditioned model’s $0.010 \rightarrow 0.149$. **Small lesions are not intrinsically unlearnable here, and Dice’s geometric size penalty is not large enough to explain the collapse — a supervised model**

absorbs both and still scores 0.64 on the smallest enhancing tumors. This is the strongest evidence in the report that the failure is specific to text-conditioned localization, and it rules out the most obvious deflationary reading of the entire project.

P' validation: the same data, split and metric under dense supervision — and how its size collapse compares

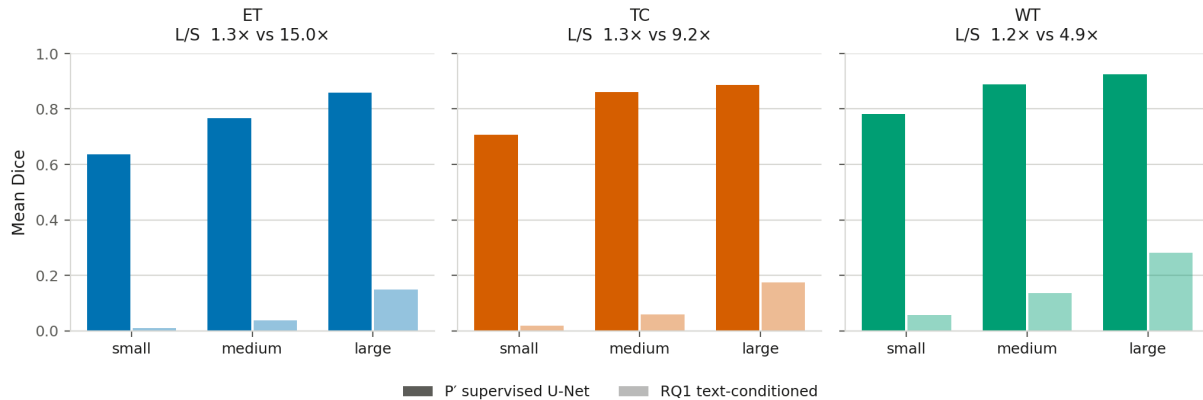


Figure 2: P' validation. Left to right: enhancing tumor, tumor core, whole tumor. Solid bars are the supervised U-Net, faded bars the text-conditioned baseline, both on the same held-out patients under the same Dice implementation. The supervised model’s large/small ratio is 1.2–1.3× against the text-conditioned 4.9–15.0×.

6.2 RQ1: Size-stratified localization

Region	Small Dice	Medium Dice	Large Dice	Large/Small ratio
ET	0.010 ± 0.009 (n=21)	0.037 ± 0.015 (n=23)	0.149 ± 0.078 (n=23)	14.9×
TC	0.019 ± 0.012 (n=27)	0.059 ± 0.022 (n=23)	0.176 ± 0.062 (n=23)	9.3×
WT	0.057 ± 0.026 (n=32)	0.137 ± 0.036 (n=21)	0.281 ± 0.049 (n=20)	4.9×

The degradation is monotonic and severe across **all three independently-defined tumor subregions**, holding on a held-out validation set never seen during training. This directly and quantitatively confirms the small-lesion grounding failure documented qualitatively in recent 3D medical VLM literature (Section 3). Absolute Dice is modest throughout (this is a lightweight ResNet-10 baseline with an unsupervised Otsu threshold, not a competitive segmentation model) — the finding is about the *relative* size-dependent collapse, not absolute segmentation quality.

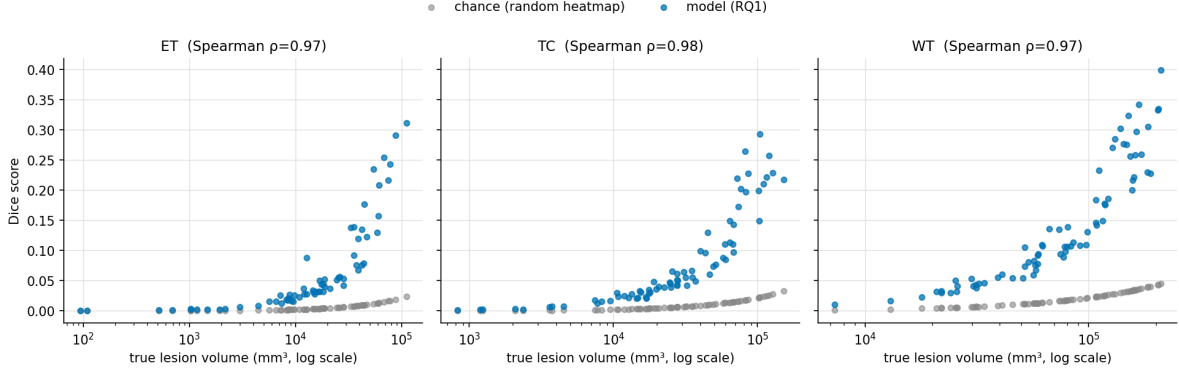


Figure 3: Dice vs. true lesion volume, model (blue) vs. a chance/random-heatmap control (gray), one panel per subregion. Both climb with volume, but the model sits well above chance throughout.

Statistical validation — is this just a Dice artifact? Dice is known to penalize small structures more harshly than large ones for *any* predictor, purely as a geometric property of the metric (a single misclassified voxel costs a tiny lesion far more of its Dice score than it costs a large one). To check whether the RQ1 collapse is a real model failure rather than this artifact, we computed a **chance baseline**: pure random-noise heatmaps run through the identical Otsu-threshold-and-Dice pipeline, on the same 73 validation patients.

The chance baseline collapses with size too — as expected, since this reflects the metric, not the model. Spearman correlation between Dice and log-volume is extremely strong for *both* the model ($\rho=0.97$ - 0.98) and chance ($\rho=0.999$ - 1.000 , i.e. almost perfectly deterministic), confirming Dice’s inherent size-dependence. The question that actually isolates the model’s behavior is the **lift over chance** (model Dice $\div$ chance Dice) at each size bin:

Region	Small lift	Medium lift	Large lift
ET	10.8×	10.7×	13.3×
TC	8.4×	8.5×	9.4×
WT	6.6×	6.7×	7.9×

This is the more careful finding: the model’s *relative* advantage over chance is roughly **constant** across size bins (within each region, small/medium/large lift are all within $\sim 25\%$ of each other) — the model isn’t disproportionately losing signal on small lesions relative to a null baseline. What collapses is **absolute** localization quality: even with a consistent ~ 7 - $13\times$ advantage over random guessing, small-lesion Dice (0.010-0.057) is nowhere near clinically usable, while large-lesion Dice (0.149-0.281), though still modest, is at least in a range serious methods report. So the honest claim is not “the model is uniquely broken on small lesions relative to its own baseline” — it’s “even a consistent relative advantage over chance is not enough to produce usable absolute localization when the target is small,” which is arguably the more clinically relevant framing anyway: a radiologist doesn’t care whether a bad prediction is bad in an absolute or relative sense.

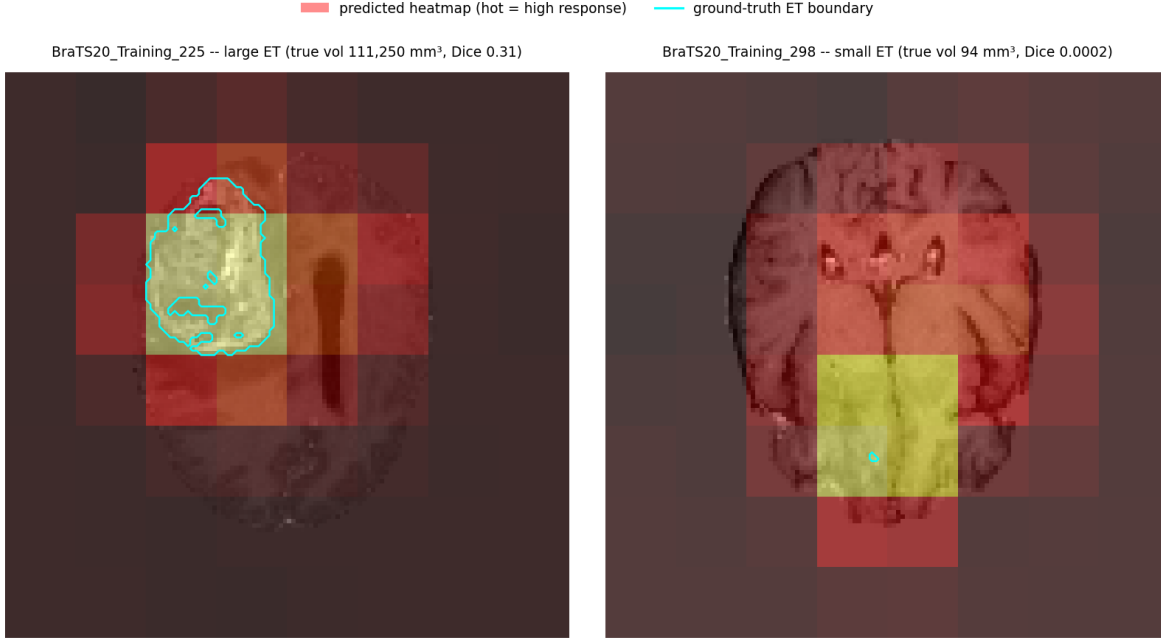


Figure 4: Qualitative example: T1ce slice with the model’s predicted heatmap (hot colormap) and the true ET boundary (cyan). Left: a large lesion, where the heatmap’s peak response aligns closely with the true tumor. Right: a small lesion, where the true tumor is a tiny dot completely swallowed by one oversized block of “high response” — a direct visual illustration of the fixed-patch-resolution problem discussed below.

The qualitative example above makes the mechanism visible: the visible blockiness in both heatmaps is the sliding-window’s fixed 32^3 receptive field. For the large lesion it happens to roughly match the tumor’s scale, so the peak response block and the true boundary line up reasonably well. For the small lesion, the entire true region fits inside a small fraction of a single response block — the window simply cannot resolve anything finer than its own size, regardless of how correct or incorrect its content judgment is.

Robustness check: does this hold on a different split? Every result up to this point uses one fixed 80/20 split (seed 0). To check this isn’t an artifact of that particular split, we retrained and re-evaluated the entire P’ baseline from scratch on two additional independent random splits (seeds 1 and 2, same hyperparameters, same protocol).

Region	Bin	seed 0	seed 1	seed 2	mean $\pm$ std
ET	small	0.0100	0.0083	0.0079	0.0087 ± 0.0009
ET	medium	0.0374	0.0375	0.0285	0.0345 ± 0.0042
ET	large	0.1488	0.1223	0.0981	0.1231 ± 0.0207
TC	small	0.0192	0.0220	0.0175	0.0196 ± 0.0019
TC	medium	0.0590	0.0738	0.0601	0.0643 ± 0.0068
TC	large	0.1756	0.2254	0.1477	0.1829 ± 0.0321
WT	small	0.0568	0.0445	0.0474	0.0496 ± 0.0053
WT	medium	0.1367	0.1220	0.1333	0.1307 ± 0.0063
WT	large	0.2809	0.2467	0.2576	0.2617 ± 0.0142

The monotonic small < medium < large pattern holds exactly in all 3 seeds, for all 3 regions (9 of 9 replications). Absolute Dice values shift somewhat between splits (expected, given different held-out patients and a modest ~ 73 -patient validation set each time), but the core relationship this project is built around is not an artifact of one particular train/val split.

6.3 RQ11: How much of the collapse is grounding failure, and how much is the threshold?

Sections 6.1-6.2 measure localization through a two-stage pipeline: a continuous similarity heatmap, then an unsupervised Otsu threshold converting it to a binary mask. Every Dice number reported so far is a property of *both* stages. The chance-level control in Section 6.2 does not separate them — a random heatmap is binarized by the same Otsu step, so a thresholding pathology would appear in the model and the control alike and cancel out of the lift ratio.

This matters because Otsu picks its cut point from each volume’s own intensity histogram, with no reference to the query or to plausible lesion size. If it emits a roughly constant-size blob regardless of the target, small lesions would score badly for a reason unrelated to text-conditioned grounding. Our result CSVs could not answer this: they never recorded how large the predicted mask actually was.

We therefore recomputed the heatmap once per (patient, region) with the identical frozen baseline and 32³/stride-16 protocol, then binarized it five ways: Otsu (the published protocol), fixed top-10%/5%/1% of voxels, and an **oracle-volume** threshold taking exactly as many voxels as the ground-truth mask contains. The last removes threshold calibration from the problem entirely and asks only how well the heatmap *ranks* voxels; it uses ground truth and is therefore a diagnostic upper bound, never a deployable method.

Otsu is badly miscalibrated, and miscalibrated in the wrong direction.

Region	Bin	GT volume	Otsu predicted	Over-prediction	% of imaged volume
ET	small	4,124 mm ³	920,036 mm ³	223.1×	10.3%
ET	large	50,716 mm ³	718,728 mm ³	14.2×	8.1%
TC	small	10,213 mm ³	1,105,667 mm ³	108.3×	12.4%
TC	large	84,708 mm ³	937,076 mm ³	11.1×	10.5%
WT	small	38,995 mm ³	1,371,023 mm ³	35.2×	15.4%
WT	large	164,364 mm ³	1,022,709 mm ³	6.2×	11.5%

Otsu returns a near-constant 8-15% of the imaged volume whatever the target’s size. Worse, the correlation between true and predicted volume is **negative in all three regions** (ET $\rho=-0.374$, $p=0.0018$; TC $\rho=-0.258$, $p=0.028$; WT $\rho=-0.475$, $p=2.2\times10^{-5}$), where a calibrated predictor would approach +1. The thresholding step assigns *larger* masks to *smaller* lesions, mechanically manufacturing part of the size effect Section 6.2 attributes to grounding.

The cost of that step is large and paired-significant in 8 of 9 bins.

Region	Bin	Otsu Dice	Oracle-volume Dice	Gain	p (Wilcoxon)
ET	small	0.0100	0.0254	2.6×	0.43 (n.s.)
ET	medium	0.0374	0.1495	4.0×	0.038
ET	large	0.1488	0.3652	2.5×	2.4×10^{-7}
TC	small	0.0192	0.0922	4.8×	0.0089
TC	large	0.1756	0.5175	2.9×	4.8×10^{-7}
WT	small	0.0568	0.2600	4.6×	2.0×10^{-6}
WT	medium	0.1367	0.5162	3.8×	9.5×10^{-7}
WT	large	0.2809	0.6218	2.2×	1.9×10^{-6}

This revises a claim made in Section 6.2. We attributed modest absolute Dice to “a lightweight ResNet-10 baseline with an unsupervised Otsu threshold” — the measurement now shows the threshold, not the backbone, carried most of that cost. Whole-tumor large-lesion Dice is 0.622 under proper binarization, not 0.281. The single exception is ET-small, the only bin where better thresholding does *not* help significantly ($p=0.43$): there the heatmap’s ranking is itself poor, so no cut point recovers it.

The size collapse survives, at reduced magnitude.

Region	L/S ratio (Otsu, as reported in 6.2)	L/S ratio (oracle-volume)	Spearman(Dice, volume) Otsu → oracle
ET	15.0×	14.4×	0.970 → 0.835
TC	9.2×	5.6×	0.978 → 0.770
WT	4.9×	2.4×	0.968 → 0.781

Enhancing tumor’s collapse is essentially entirely genuine; tumor core’s and whole tumor’s are roughly half thresholding artifact. The honest headline is therefore **2.4-14.4× degradation after controlling for binarization**, not 5-15×. The relationship remains monotonic, strongly positive, and present in every region under every one of the five binarization rules tested — it is the magnitude, not the existence, that was overstated.

A threshold-independent metric isolates where the real failure lives. Dice conflates “did the model point at the lesion” with “did it draw the right boundary.” We therefore also report the **pointing game** (Zhang et al., 2018): is the highest-response location inside the true mask? It is free of Dice’s geometric size penalty. Its chance level still scales with target size, so we compare against a per-bin chance baseline (mean GT voxels ÷ total voxels) with an exact binomial test.

First, a correction to how that peak is located. A sliding window of stride s gives every voxel the mean of the windows covering it, and all voxels inside the same s^3 block are covered by an identical window set — so the heatmap is *piecewise-constant over s^3 blocks*, not smooth. At our published $32^3/\text{stride-16}$ protocol that block is $16^3 = 4096$ voxels, roughly 30 mm across, and `argmax` silently returns the block’s **corner** in array order rather than anything peak-like. We verified this directly: the median count of voxels tied at the maximum is exactly 4096 at stride 16 and exactly 512 at stride 8, matching s^3 in both cases. We therefore locate the peak at the **centroid of the tied-maximum plateau** and report both rules below, since the naive rule biases hit rates down and distances up.

Region	Bin	n	Hits (corrected)	Hit rate	Chance	Lift	p	Median distance
ET	small	21	0	0.000	0.0005	0.0×	1.00	23.7 mm
ET	medium	23	3	0.130	0.0018	74.5×	9.3×10^{-6}	9.4 mm
ET	large	23	8	0.348	0.0057	61.2×	4.9×10^{-13}	2.9 mm
TC	small	27	3	0.111	0.0011	97.1×	4.3×10^{-6}	12.0 mm
TC	medium	23	8	0.348	0.0035	99.0×	1.1×10^{-14}	1.9 mm
TC	large	23	16	0.696	0.0095	73.3×	9.9×10^{-28}	0.0 mm
WT	small	32	17	0.531	0.0044	121.6×	4.1×10^{-32}	0.0 mm
WT	medium	21	18	0.857	0.0105	81.9×	2.9×10^{-33}	0.0 mm
WT	large	20	17	0.850	0.0184	46.2×	3.5×10^{-27}	0.0 mm

The correction matters most where the model was already doing well: whole-tumor small goes from 6/32 hits to 17/32, and tumor-core large from 13/23 to 16/23. **The headline claim is unaffected: enhancing-tumor small remains 0 hits in 21 patients under both rules.** The old first-argmax numbers are retained in the analysis log for comparison.

Chance-corrected lift is roughly **uniform at 46-122× across every region and size bin** — echoing Section 6.2’s finding that lift over chance is approximately constant, but now with a metric that has no built-in size bias. The exception is stark and specific: **for small enhancing tumor the model is at exact chance, 0 hits in 21 patients, with its peak response a median 23.7 mm from the nearest lesion voxel.** That is not a boundary-delineation failure; the model is not pointing at the lesion at all. This is the sharpest statement of the failure this project can make, and it is confined to one region×bin rather than being the smooth monotonic collapse the Dice framing suggests. Section 7.12 returns to this bin and shows it is the one place the smaller-window intervention genuinely helps.

One deployable consequence. The fixed top-1% threshold (pct99) needs no ground truth and beats Otsu everywhere, by 3-4× for ET (0.0436/0.1774/0.3672 vs 0.0100/0.0374/0.1488). For ET it even beats the oracle-

volume threshold in all three bins — when a heatmap’s ranking is unreliable, a mask larger than the true lesion is Dice-optimal, because concentrating into exactly the right *number* of voxels is only rewarded if they are the right *ones*. Replacing Otsu with a fixed percentile is a free protocol improvement, and unlike the oracle it is usable at inference time.

7. Ablation Studies

RQ1 established that the failure is real. This section systematically ablates candidate explanations and fixes — text-side conditioning, inference-side window size, and training-side retraining — to find out whether the failure is linguistic, architectural, or trainable away. Figure below previews the full decision tree and how each ablation’s outcome motivated the next.

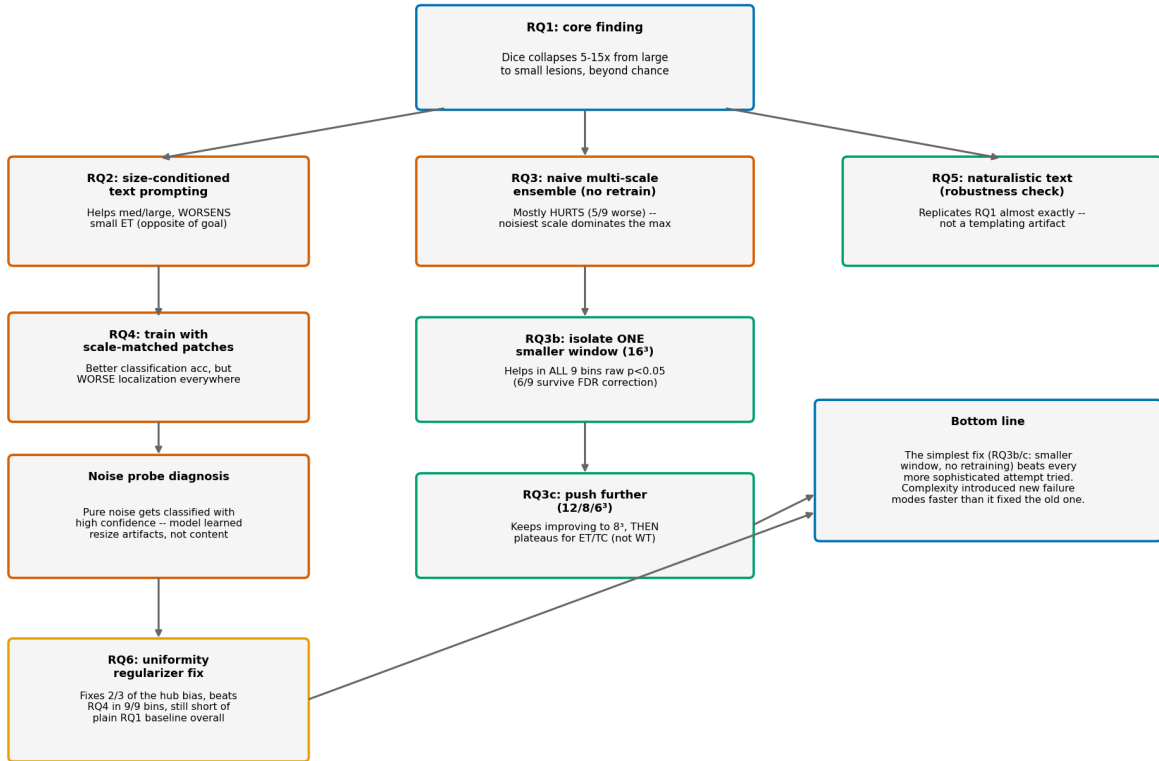


Figure 5: Roadmap of every ablation attempted, from RQ1’s core finding through RQ6, with each box’s one-line outcome. Green = the intervention helped, red = it hurt or introduced a new problem, orange = partial success.

7.1 RQ2: Size-conditioned prompting mitigation

Mechanism: at training time, each ET/TC/WT patch is labeled with its lesion’s *true* size bin (known from ground truth) and trained against a size-specific text description (10-way classification: 3 regions $\times$ 3 sizes + NONE). At evaluation time, since true size is unknown a priori, we query with all three size-phrasings per region and take the voxel-wise maximum response (a deployable ensemble, not test-time label leakage).

Training: validation accuracy on the finer-grained 10-way task reached 0.552 (chance = 0.10) after 30 epochs.

Region	Bin	RQ1 (baseline)	RQ2 (size-conditioned)	Δ
ET	small	0.010	0.008	−0.002
ET	medium	0.037	0.038	+0.001

Region	Bin	RQ1 (baseline)	RQ2 (size-conditioned)	Δ
ET	large	0.149	0.191	+0.042
TC	small	0.019	0.027	+0.008
TC	medium	0.059	0.095	+0.036
TC	large	0.176	0.252	+0.076
WT	small	0.057	0.059	+0.002
WT	medium	0.137	0.147	+0.010
WT	large	0.281	0.321	+0.040

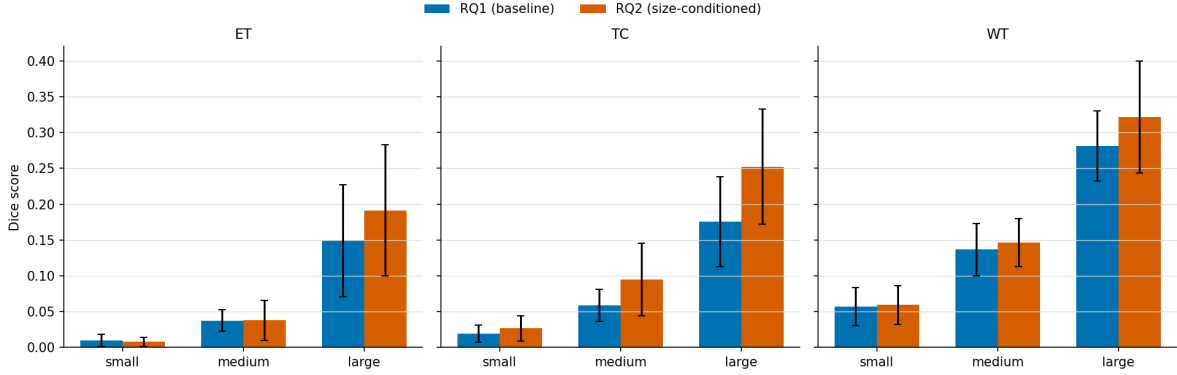


Figure 6: RQ1 (baseline) vs RQ2 (size-conditioned) Dice by region and size bin, error bars = 1 std.

Statistical validation. A paired Wilcoxon signed-rank test (matched by patient, RQ1 vs RQ2 Dice) at each region $\times$ size bin gives a precise picture, and it’s more nuanced than the raw means alone suggest:

Region	Small bin	Medium bin	Large bin
ET	significantly worse ($p=0.010$)	no significant change ($p=0.23$)	significantly better ($p<0.0001$)
TC	significantly better, tiny effect ($p<0.0001$, $+0.008$)	significantly better ($p<0.0001$, $+0.036$)	significantly better ($p<0.0001$, $+0.076$)
WT	significantly better, tiny effect ($p=0.017$, $+0.002$)	not significant ($p=0.050$, borderline)	significantly better ($p=0.004$)

Finding: the mitigation did not work for its intended purpose, and the effect is not uniform across regions. For ET — the smallest and hardest region, and arguably the one that matters most clinically — size-conditioned prompting made small-lesion localization *significantly worse* ($p=0.010$), the opposite of its goal. For TC and WT, small-bin Dice did improve with statistical significance, but the effect sizes are clinically negligible ($+0.008$ and $+0.002$ respectively) — real, but not meaningful. In relative terms the large/small gap for ET actually *widened* ($14.9\times \rightarrow 23.9\times$). Meanwhile every region shows a real, often large, improvement at the medium/large end (TC medium $+61\%$ relative, TC large $+43\%$ relative, both $p<0.0001$). So the mitigation is doing something — it is reliably better at medium/large lesions — but that something is not solving, and for the hardest region actively worsens, the specific small-lesion problem it was designed to fix.

Why, most likely: the sliding-window patch is a fixed 32^3 voxels, which at this resampled resolution (128^3 from an original $\sim 240\times 240\times 155$ grid) spans roughly $60\times 60\times 39\text{mm}$ physically. A “small” ET lesion can be as little as 32mm^3 — a handful of voxels — meaning even a perfectly-worded “small lesion” text query is matched against a window enormously larger than the lesion itself. The lesion’s signal is diluted by surrounding normal tissue inside the patch regardless of what text conditions the query. This points to a **resolution/architecture bottleneck, not a language bottleneck** — you cannot prompt your way out of a fixed receptive field that’s larger than the target.

7.2 RQ3: Does the receptive field actually explain it? (naive multi-scale windowing)

RQ2’s explanation is a claim, not yet a test. If a fixed 32^3 receptive field really is the bottleneck, then evaluating the *same, frozen* RQ1 model at a smaller physical window (resized to the model’s 32^3 input before encoding) should help, with no retraining at all. We swept three window sizes (16^3 , 32^3 , 64^3 , each with stride = window/2) and combined them via voxel-wise maximum.

Region	Small bin	Medium bin	Large bin
ET	not significant (p=0.43)	significantly worse (p=0.0001)	significantly worse (p=0.016)
TC	not significant (p=0.40)	not significant (p=0.12)	significantly worse (p=0.014)
WT	significantly better (p=0.033)	significantly worse (p=0.001)	significantly worse (p=0.006)

Naive multi-scale ensembling mostly hurts. One of nine bins improved; five got significantly worse; three showed no significant change. (All of these hold after BH-FDR correction across the full test family — q-values track the raw p-values closely here.) This seems to contradict RQ2’s explanation — but a quick smoke test on a single patient hinted at why it might not: the 16³ window alone produced a much higher peak similarity score (0.259) than the 32³ baseline (0.061), while 64³ produced uniformly *negative* scores everywhere (the model was never trained to interpret a resized-down 64³ crop, so it just never fires). Voxel-wise max across scales means that wherever *any* scale spuriously spikes, that noise wins — and a scale the model can’t interpret meaningfully (64³) is exactly the kind of scale that produces uncalibrated, noisy responses.

7.3 RQ3b: Isolating the effect — one smaller window, no ensembling

To separate “does a smaller receptive field help” from “does naively combining multiple scales help,” we re-ran evaluation using *only* the 16³ window (no ensembling with 32³ or 64³), still on the same frozen RQ1 model, still no retraining.

Region	Small Dice	Medium Dice	Large Dice
ET	0.011 (+11%)	0.039 (+6%)	0.160 (+8%)
TC	0.023 (+21%)	0.069 (+17%)	0.198 (+13%)
WT	0.072 (+26%)	0.156 (+13%)	0.343 (+22%)

(percentages are relative improvement over the RQ1 32³ baseline)

All 9 region×bin combinations improved at raw p<0.05 — but this needs the multiple-comparisons correction applied honestly. After BH-FDR correction across the full accumulated test family, **TC’s and WT’s improvements survive (6/6, q<0.05 throughout)**, but **ET’s three bins do not** (q=0.052, 0.052, 0.064 — just above the corrected threshold). ET is also the region the report has repeatedly flagged as the smallest, hardest, and clinically most important — so the one region where this fix matters most is exactly the one where the statistical evidence is weakest. The honest claim is: a smaller window robustly helps TC and WT localization at every size, and probably helps ET too, but that specific claim doesn’t clear a properly corrected significance bar with n=21-23 patients per bin.

Where the win is real (TC, WT), it also isn’t free computationally. The 16³/stride-8 sweep does 3,375 forward passes per volume versus the 32³/stride-16 baseline’s 343 — a 9.8× increase — and matches in wall-clock: the RQ1 evaluation job ran in 2m47s, the RQ3b job in 19m17s, roughly 7×. “No retraining needed” is accurate; “free” is not, and we shouldn’t have said it that way.

Even with those caveats, the large/small *ratio* barely moves either way (ET 14.9×→14.4×, TC 9.3×→8.6×, WT 4.9×→4.8×) — this is a real lift to absolute quality for two of three regions, not a fix for the underlying relative size gap. Going from “our own explanation sounds plausible” to “we tested it, and it holds for 6 of 9 bins after correcting for multiple comparisons” is still real progress over what RQ3 alone showed.

7.4 RQ3c: How far does “smaller is better” go?

RQ3b only compared one alternative (16³) against the 32³ baseline. To see whether the trend continues, reverses, or plateaus, we extended the same isolated-window evaluation (no ensembling, no retraining, same frozen RQ1

model) to three smaller sizes: 12^3 (stride 6), 8^3 (stride 8), and 6^3 (stride 6; the latter two use non-overlapping tiling rather than the 50%-overlap convention used elsewhere, for compute reasons — flagged here rather than left implicit).

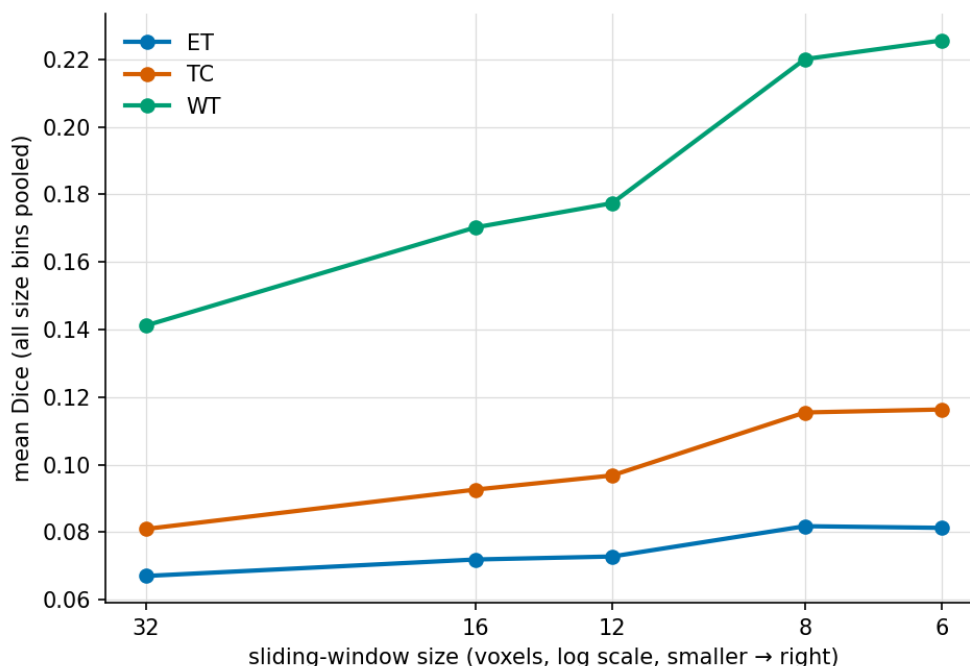


Figure 7: Mean Dice (pooled across size bins) vs. sliding-window size, one line per region. The trend from RQ3b continues through $12^3/8^3$, then plateaus for ET/TC at 6^3 while WT keeps climbing.

The trend **continues in the same direction through 8^3 , with no plateau or reversal**:

Region	32^3 (baseline)	16^3	12^3	8^3	6^3
ET large	0.149	0.160	0.161	0.177	0.177
TC large	0.176	0.198	0.208	0.242	0.245
WT large	0.281	0.343	0.360	0.426	0.437

More importantly, re-running the full BH-FDR correction across all tests accumulated up to this point, **ET’s statistical picture actually improves as the window shrinks further**: at 12^3 , 2 of ET’s 3 bins now survive correction (only ET-large is borderline, $q=0.052$); at 8^3 , **all 3 of ET’s bins survive**, along with every other region and bin (18/18 at this window size and the one below it). The region that couldn’t clear a corrected significance bar at 16^3 becomes the region with the cleanest signal at 8^3 .

Pushing one step further to 6^3 , the trend plateaus for ET and TC specifically. A paired Wilcoxon test between 8^3 and 6^3 shows no significant difference for any ET or TC bin (all 6 bins n.s., $p=0.13-0.95$), but WT continues to improve significantly in all 3 bins ($p<0.05$ throughout, including $p=0.0006$ for WT small). A plausible reason: WT lesions are the largest of the three regions by a wide margin (median $\sim 90,740\text{mm}^3$ vs. ET’s $\sim 16,971\text{mm}^3$ and TC’s $\sim 33,808\text{mm}^3$), so even an 8^3 or 6^3 window is still comparatively small relative to WT’s typical physical extent, leaving more room to benefit from further shrinking, while ET and TC — already much smaller lesions — have apparently reached the point where the window is no longer the binding constraint. This directly answers the question posed after RQ3c’s first pass: the “smaller is better” trend does have a floor, and it isn’t the same floor for every region.

7.5 RQ4: Training with scale-matched patches

RQ3b changes only *evaluation*. The more ambitious version of the same idea is to also *train* with patches whose physical crop size matches the declared size bin (small→16³, medium→32³, large→64³, each resized to the canonical 32³ input), combined with RQ2’s size-conditioned text (10-way classification), then evaluate with each size-phrasing queried at its matched scale and combined via max.

Training reached a *better* classification accuracy than RQ2 (0.668 vs. 0.552 best val accuracy on the analogous 10-way task) — the model clearly learned to use scale as a signal. But localization got **significantly worse than the RQ1 baseline in all 9 bins (p<0.0001 throughout, and all 9 survive BH-FDR correction)**. We also directly verified the comparison to RQ2 rather than asserting it: a paired test confirms RQ4 is **significantly worse than RQ2 in all 9 bins as well (p<0.0001 throughout, all surviving correction)**.

Region	Small Dice	Medium Dice	Large Dice
ET	0.005	0.018	0.064
TC	0.012	0.036	0.098
WT	0.048	0.106	0.178

Better classification accuracy did not transfer to better localization — and we tested why, rather than just speculating. The hypothesis: since each size-conditioned class was trained on patches resized in a systematically different direction (16³ crops upsampled/blurred, 64³ crops downsampled/detail-lost), the model might key off *resize-interpolation artifacts* correlated with the size label rather than genuine tumor content. We tested this directly: we fed **pure random Gaussian noise** — zero real anatomical content — through the identical three crop-then-resize pipelines used in training, and measured the trained RQ4 model’s similarity between the resulting image embeddings and each size-conditioned text embedding.

Noise pipeline (crop size)	Mean similarity to its own matching size-text
“small” (16 ³ →32 ³)	−0.401 ± 0.010
“medium” (32 ³ native)	+0.026 ± 0.010
“large” (64 ³ →32 ³)	+0.149 ± 0.008

A one-way ANOVA across the three noise groups gives $F=59,672$, $p\approx 4\times 10^{-251}$ — an essentially complete, non-overlapping separation, on pure noise with no tumor present. **This confirms the model is not relying on real content for a substantial part of its size signal.** The mechanism is more specific than we first guessed, though, and worth reporting precisely rather than rounding it off to the original hypothesis: it is not that each pipeline gets cleanly fingerprinted to its own label. Every noise pipeline — including “small” and “medium” — scores *highest* against the **“large”** text embedding (small-pipeline noise: −0.400 vs. “small” text, but +0.015 vs. “large” text; medium-pipeline noise: +0.023 vs. “medium” text, but +0.126 vs. “large” text). This looks like a generic bias toward the “large” class dominating almost regardless of input — an embedding-space hub or degenerate solution — rather than genuine per-scale artifact recognition. Either way, the conclusion is the same: RQ4’s size classification is not reliably grounded in real tumor content, which is a sufficient and now directly-verified explanation for why its improved training-time accuracy failed to produce better localization. Combined with RQ3’s finding that max-ensembling across scales amplifies whichever scale is noisiest, RQ4 appears to inherit both problems at once.

7.6 RQ6: Can the embedding hub actually be fixed?

RQ4 diagnosed a real problem (the “large”-class hub) rather than solving one. We attempted an actual fix: retrain the identical scale-matched setup with an added **uniformity regularizer** — a loss term that directly penalizes high pairwise cosine similarity among the projected text class embeddings, pushing them apart on the hypersphere so no single class can act as a generic attractor regardless of input content.

It worked, partially, and we verified each part of that claim rather than asserting it:

The embeddings actually separated. Mean pairwise cosine similarity among the text projections fell from ~ 0.97 (matching RQ4’s near-total collapse) to **-0.10** within a few epochs of training, and stayed there. Re-running the exact noise probe from RQ4 confirms this is a real, not just numerical, change: **2 of 3 noise pipelines now correctly prefer their own matching label** (medium-pipeline noise scores highest on “medium” text, large-pipeline noise scores highest on “large” text), up from 0 of 3 in RQ4. Only the smallest-scale pipeline still shows residual bias toward “large.”

Localization genuinely improved over RQ4 — verified with paired tests, not eyeballed. RQ6 beats RQ4 in **all 9 of 9 region $\times$ bin combinations**, every one significant ($p < 0.01$, most $p < 0.001$).

Region	RQ4 small	RQ6 small	RQ4 medium	RQ6 medium	RQ4 large	RQ6 large
ET	0.005	0.005	0.018	0.020	0.064	0.070
TC	0.012	0.016	0.036	0.046	0.098	0.120
WT	0.048	0.061	0.106	0.137	0.178	0.226

But RQ6 still falls short of the plain RQ1 baseline in most bins. Paired against RQ1, RQ6 is significantly *worse* in all of ET and TC (7 of 9 bins), and only statistically tied (not better) in WT small/medium, worse again in WT large. Fixing the embedding collapse was necessary but not sufficient.

A follow-up isolates why, and the answer reverses our own expectation. We hypothesized RQ4/RQ6’s other diagnosed problem — RQ3’s finding that max-ensembling across scales amplifies noise — was still holding RQ6 back, and that evaluating with *only* the correctly-matched single scale per patient (no ensembling) would do better. We tested this directly (using the true size bin as an oracle purely to isolate the mechanism, not as a deployable evaluation) and found the **opposite**: the single-scale version is significantly *worse* than RQ6’s own 3-way ensemble in 8 of 9 bins, dramatically so for large lesions (roughly half the Dice: e.g. WT large $0.226 \rightarrow 0.108$).

This means ensembling across scales is not inherently harmful, as RQ3 seemed to show — it depends entirely on whether the underlying model was actually trained to produce calibrated responses at each scale. RQ1 was trained only at 32^3 and evaluated out-of-distribution at $16^3/64^3$ in RQ3, where ensembling amplified noise. RQ4/RQ6 were specifically trained across all three scales, and for them, ensembling captures genuine complementary information — most likely because a single window, even at the “correct” scale, cannot cover a large lesion’s full spatial extent, while sweeping multiple scales and taking the max lets different parts of a large lesion register at whichever scale best captures them.

7.7 RQ5: Does the finding survive naturalistic language?

All of RQ1/RQ2’s text was templated, textbook-style description. To rule out the entire size-collapse finding being some artifact of that specific phrasing, we rewrote the four base descriptions in naturalistic, radiology-report style (hedged, varied syntax — e.g. “*Post-contrast T1-weighted images demonstrate irregular nodular enhancement, favoring viable, high-grade tumor tissue*” rather than “*Enhancing tumor: region of active contrast enhancement...*”), retrained from scratch, and re-ran the identical RQ1 evaluation protocol.

Result on this single run: **essentially no difference.** 8 of 9 region $\times$ bin comparisons show no significant difference from the original templated-text RQ1 (paired Wilcoxon, all $p > 0.07$), and Spearman correlation between Dice and log-volume is nearly identical ($\rho = 0.958$ templated vs. 0.959 naturalistic). The one significant difference (TC large, $p = 0.0035$) is a small improvement.

This conclusion is partly revised in Section 7.10. Retrained under three seeds, naturalistic text is consistently *better* than templated ($+0.040$ pooled Dice, positive in all three runs), so “no difference” was a seed-0 artifact and this is a weaker robustness check than it appears here. What survives is the part that matters for the paper: the size collapse persists under naturalistic text in every seed, so the finding is not a templating artifact.

7.8 RQ7: Does the text encoder contribute anything at all?

Every ablation so far manipulated *what the text says*. RQ7 asks a blunter question: does it matter *what produces the embedding*? We retrained the identical architecture and protocol under four substituted text conditions, changing nothing else:

- **BERT-base** — general-domain rather than biomedical pretraining.
- **Random-init BERT (frozen)** — the PubMedBERT architecture with randomly initialized weights, never trained. Preserves BERT’s embedding *geometry* but carries no learned semantics.
- **4 random orthonormal vectors** — pure class identifiers. No language, no geometry inherited from any encoder.
- **Random vectors at PubMedBERT’s geometry** — random vectors resampled to match PubMedBERT’s anisotropic covariance structure. Semantically empty, but geometrically identical to the real thing.

The last two are the decomposition that makes this informative: comparing PubMedBERT against *random vectors at its own geometry* isolates the contribution of **meaning** with geometry held fixed, while comparing orthonormal against anisotropic random vectors isolates the contribution of **geometry** with semantics held fixed at zero.

The single-seed result looked dramatic and was largely wrong. Pooled over all regions and bins, patient-paired tests reported random-init BERT *beating* PubMedBERT by +0.038 Dice at $p \approx 6 \times 10^{-35}$, and BERT-base beating it by +0.015 at $p \approx 4 \times 10^{-15}$. Those p-values are not defensible. Each condition was trained exactly once, so the unit of randomization is the *training run*, not the patient; treating 213 patients from one run as 213 independent observations is pseudo-replication. The check that exposed it: retraining the *identical* PubMedBERT model under three seeds moves pooled Dice by 0.0086 on its own — comparable to several of the effects being claimed.

Retrained under three seeds each, almost nothing survives.

Text condition	seed 0	seed 1	seed 2	mean Δ	verdict
BERT-base (general domain)	+0.0151	−0.0378	+0.0078	−0.0050	sign flips → noise
Random-init BERT (frozen)	+0.0379	+0.0283	−0.0014	+0.0216	sign flips → noise
Random vectors, PubMedBERT geometry	−0.0072	+0.0370	+0.0085	+0.0128	sign flips → noise
4 random orthonormal vectors	−0.0140	−0.0103	−0.0020	−0.0087	consistent, 2.0× noise floor

Noise floor (baseline retrained, 3 seeds) = 0.0044 pooled Dice.

Three findings follow, and the first two are uncomfortable.

Biomedical pretraining is worth nothing measurable here. Swapping PubMedBERT for general-domain BERT-base produces an effect that changes sign across training runs. Whatever domain knowledge PubMedBERT encodes about enhancing tumor and peritumoral edema, this pipeline does not use it.

Neither is language. Replacing the encoder entirely with random vectors — no text, no tokenizer, no pretraining — is also indistinguishable from noise, provided those vectors carry PubMedBERT’s anisotropic geometry. The one condition that *does* reliably underperform is the one that discards that geometry (orthonormal vectors, −0.0087, consistent across all three seeds). The decomposition is therefore stark: **the semantic content of the text contributes nothing detectable, while its embedding geometry contributes a small but reproducible amount.** Section 3 noted that our four class descriptions sit at ~ 0.99 pairwise cosine similarity; RQ7

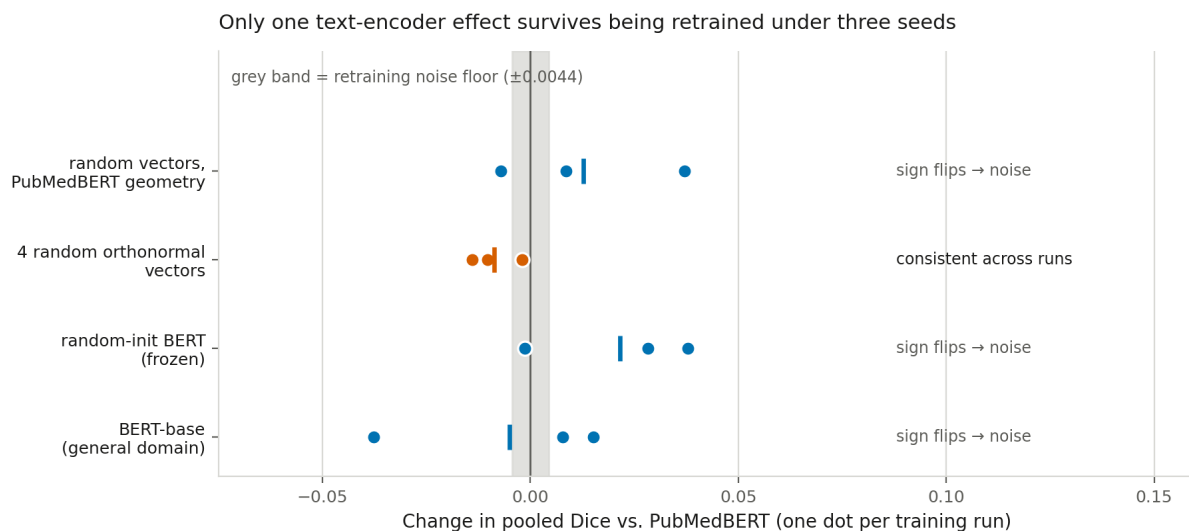


Figure 8: RQ7 text-encoder ablations. Each dot is one independent training run; the vertical tick is the mean of the three. The grey band is the retraining noise floor, measured by retraining the *identical* PubMedBERT baseline under three seeds. Only the orthonormal-vector condition (orange) keeps its sign across all three runs; the other three change direction from run to run despite within-run p-values as small as 6×10^{-35} .

shows the trainable projection head is not merely doing *most* of the discriminative work, it is doing essentially *all* of it, and it works about as well starting from arbitrary vectors as from meaning.

The core finding is encoder-independent. The large/small Dice ratio stays in the same range under every text condition and every seed (ET 10.6–18.2 $\times$, TC 7.3–17.6 $\times$, WT 4.6–5.6 $\times$). The small-lesion collapse is not a property of PubMedBERT; it survives replacing the language model with noise.

7.9 RQ8: Is the query functioning as language, or as a class identifier?

RQ7 shows the encoder can be replaced. RQ8 asks the complementary, more mechanistic question directly of the *trained* baseline: does it respond to linguistic structure at all? We built four manipulations of each region description and re-ran the frozen RQ1 model on all of them — evaluation only, no retraining:

- **negated** — the same clinical content under explicit negation (“no enhancing tumor is seen”). Preserves nearly every content word while inverting the meaning.
- **shuffled** — word order destroyed by a fixed-seed permutation. Holds the bag of words exactly constant, removes only syntax.
- **swapped** — the sentence frame kept, the region’s head term replaced by a different region’s.
- **generic** — contentless filler naming no anatomy or pathology at all (“a region of tissue”). The floor condition.

The “original” condition reproduces `rq1_localization_scores.csv` exactly, which is an automatic end-to-end correctness gate on the script.

The model barely notices any of it.

Manipulation	Cosine to original, <i>after</i> the trained projection	Heatmap Spearman ρ vs. original
negated	0.940 – 0.962	0.561 – 0.939
shuffled	0.962 – 0.974	0.753 – 0.996
swapped	0.954 – 0.981	0.793 – 0.965
generic	0.940 – 0.947	0.777 – 0.924

Every manipulation stays above 0.94 cosine to the original query *after* the trained text head — the component supposedly responsible for making these classes discriminable. The projection does not separate “enhancing tumor” from “no enhancing tumor” from “a region of tissue” in any strong sense.

Contentless and wrong-referent text sometimes localizes better than the real description. For whole

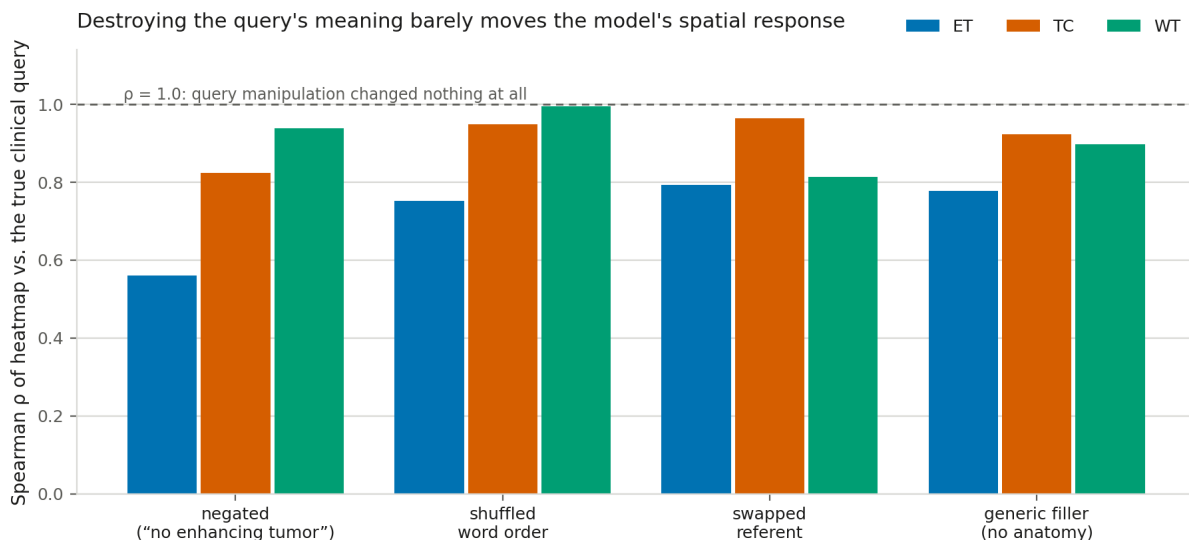


Figure 9: RQ8 compositionality probes. Spearman correlation between each manipulated query’s heatmap and the true clinical query’s, averaged over held-out patients. A model that composed meaning would drop sharply under negation and under word-order destruction; instead every bar sits between 0.56 and 1.00, and shuffling word order leaves whole-tumor behaviour essentially unchanged ($\rho=0.996$).

7.10 Do the ablation conclusions replicate across training runs?

Section 6.2 replicated the *core* finding across three splits, but every ablation conclusion above rested on one training run each — exactly the trap RQ7 exposed. We retrained and re-evaluated RQ2, RQ4, RQ5 and RQ6 under seeds 0/1/2.

Ablation	seed 0	seed 1	seed 2	mean Δ	region×bin combinations replicating in all 3
RQ2 (size-conditioned text)	+0.0227	+0.0098	+0.0258	+0.0194	4 / 9
RQ4 (scale-matched)	-0.0376	-0.0466	-0.0109	-0.0317	6 / 9
RQ5 (naturalistic text)	+0.0035	+0.0528	+0.0641	+0.0401	6 / 9
RQ6 (uniformity fix)	-0.0231	-0.0378	-0.0295	-0.0301	7 / 9

All four hold their *direction* in all three runs, so the headline verdicts — RQ2 helps on average, RQ4 and RQ6 hurt — are not split artifacts. Three qualifications follow, and one is a correction.

A correction to Section 7.7. RQ5 was reported as showing “essentially no difference” from templated text. That was a seed-0 artifact. Across three runs naturalistic text is consistently *better* (+0.040 pooled, positive in all three seeds, 9× the noise floor), driven by TC and WT which replicate 3/3 in every size bin. The $n=3$ one-sample t -test is $p=0.16$ and therefore not significant on its own — with three runs it has almost no power — so the honest statement is: **consistent in direction and large relative to retraining noise, but not formally significant at this replication count.** What does *not* change is the conclusion that mattered: the size collapse persists under naturalistic text (L/S ratios 11.6–16.2× ET, 6.4–9.6× TC, 4.1–4.9× WT), so the finding is still not a templating artifact. RQ5 is a weaker robustness check than claimed and a stronger positive result than claimed.

RQ2’s per-bin story is the least stable. Only 4 of 9 combinations replicate in all three seeds. Specifically, the headline claim that size-conditioned prompting *worsens* small enhancing tumor holds in 2 of 3 seeds (−0.0021, −0.0015, +0.0042), and the whole-tumor small bin flips outright (1/3). The robust part of RQ2 survives and

212 of 213 patients bit-for-bit (the one exception differs by 5×10^{-4} Dice, from a tie in Otsu’s 256-bin histogram), which is an end-to-end correctness gate on the generalization.

The Dice win exists under Otsu and under nothing else.

Binarization rule	Window 32^3	Window 8^3	Δ	p	8^3 wins?
Otsu (the published protocol)	0.0972	0.1407	+0.0435	3.8×10^{-34}	yes
pct90	0.1034	0.1020	-0.0014	3.7×10^{-18}	no
pct95	0.1774	0.1738	-0.0036	3.9×10^{-08}	no
pct99 (deployable)	0.2968	0.2574	-0.0394	6.4×10^{-4}	no
oracle-volume (diagnostic)	0.3085	0.2652	-0.0433	1.0×10^{-2}	no

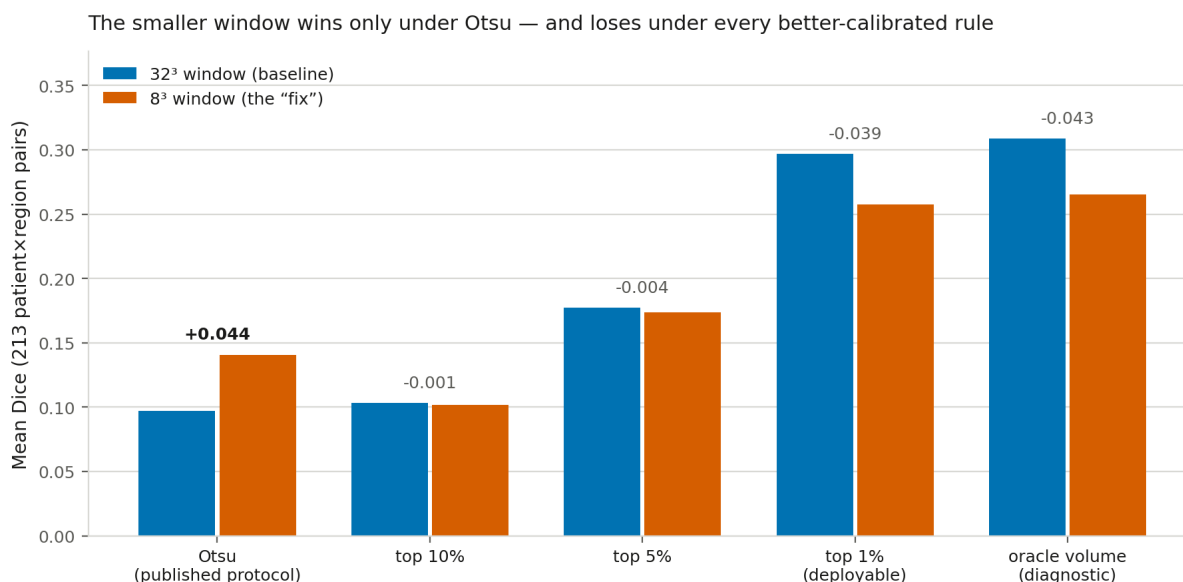


Figure 10: RQ12. The same two models, the same patients, the same Dice implementation — scored under five binarization rules. The 8^3 window’s advantage exists only in the leftmost group, the threshold this project standardized on; under the deployable top-1% rule and the oracle-volume diagnostic it reverses.

Pooled over all 213 patient×region pairs, the 8^3 window beats 32^3 by +0.044 Dice under Otsu and *loses* under every other rule, by a margin as large as the win under the two rules that best reflect heatmap quality. The pct90/pct95 deltas are statistically significant but negligible in magnitude ($|\Delta| < 0.004$) and we do not read them as substantive; pct99 and oracle-volume are both substantive and both negative.

This overturns Section 7.11’s previous bottom line. The smaller window was not localizing better. It was producing a heatmap whose intensity histogram happens to interact more favourably with Otsu’s between-class-variance criterion. Change the binarizer to anything better calibrated and the advantage inverts. The published +0.044 was measuring the coupling between a model and a bad thresholder, not grounding quality — and no amount of paired testing or FDR correction would have caught it, because every arm shared the same confounded metric.

The pointing game splits sharply by region, and rescues one specific claim.

Region	Bin	32^3 hit rate	8^3 hit rate	Change
ET	small	0.000	0.143	+0.143
ET	medium	0.130	0.391	+0.261

Region	Bin	32 ³ hit rate	8 ³ hit rate	Change
ET	large	0.348	0.565	+0.217
TC	small	0.111	0.000	−0.111
TC	medium	0.348	0.043	−0.304
TC	large	0.696	0.087	−0.609
WT	small	0.531	0.375	−0.156
WT	medium	0.857	0.571	−0.286
WT	large	0.850	0.850	±0.000

For enhancing tumor — the smallest, clinically hardest region — the smaller window improves pointing accuracy in every size bin. **Most notably it partially repairs the single sharpest failure in this report: ET-small goes from 0 of 21 patients to 3 of 21, which is 309× its chance level ($p=1.3\times 10^{-7}$) rather than exactly at chance.** For tumor core and whole tumor it makes pointing dramatically worse, collapsing TC-large from 0.696 to 0.087.

So the smaller window is not a general improvement and not a general failure. It trades away localization of large structures to gain localization of small ones — which is a coherent consequence of shrinking the receptive field, and is exactly the trade the project was looking for, just far narrower than the Dice numbers implied. The correct claim is: **an 8³ query window measurably improves grounding for enhancing tumor, including the bin that was previously at chance, while degrading it everywhere else and losing on Dice under every well-calibrated threshold.**

7.12 Summary across ablations

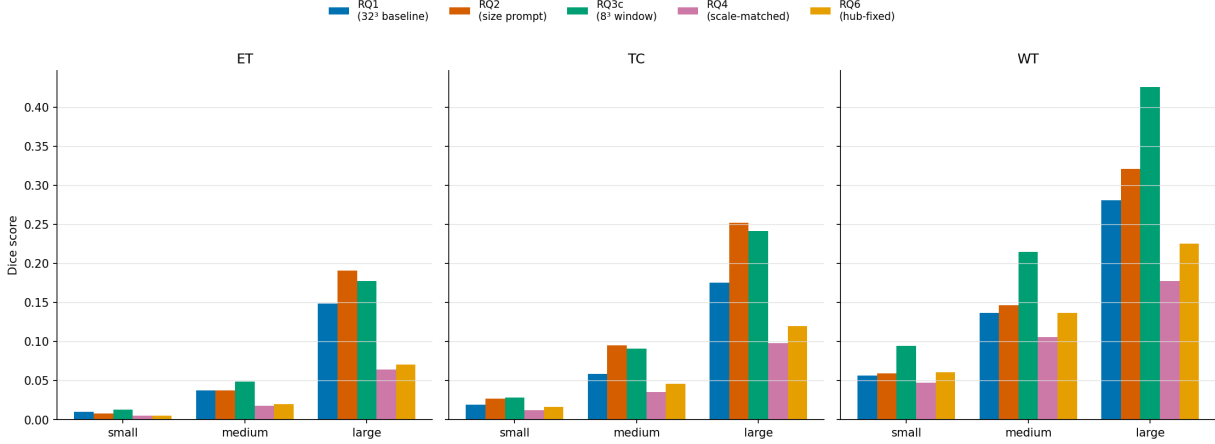


Figure 11: Leaderboard: every method’s mean Dice, one panel per region, bars grouped by size bin. Note that these are Otsu-thresholded scores; Section 7.11 shows the RQ3c ranking does not survive a better binarizer.

The honest bottom line, after Section 7.11: this project did not find an intervention that improves small-lesion grounding across the board. The candidate that appeared to — evaluating the frozen model at a smaller query window — turned out to win only under the specific threshold the whole project had standardized on, and to lose under better-calibrated ones. Every other attempt (text-side size conditioning in RQ2, scale-matched retraining in RQ4, embedding-collapse repair in RQ6) produced genuine, verifiable partial successes on its own terms, and none surpassed the plain baseline.

Two positive results do survive. First, the smaller window genuinely improves *pointing* accuracy for enhancing tumor in all three size bins, including lifting the previously-at-chance ET-small bin to 309× chance — a narrow but real fix to the report’s sharpest failure. Second, replacing Otsu with a fixed top-1% threshold is a free, deployable protocol improvement worth 2-4× Dice on its own (Section 6.3), and it requires no retraining and no ground truth.

The most valuable output of this project is therefore diagnostic rather than remedial: a set of controls — chance baselines, noise probes, cross-seed replication, pipeline decomposition, metric triangulation, and metric well-definedness — that between them caught four claims this work would otherwise have reported as findings. Sections 7.8-7.11 each overturned or substantially qualified a conclusion the earlier sections had already written down.

8. Diagnostic Methodology

The ablation studies above rely on four recurring diagnostic tools, used consistently across every experiment rather than invented ad hoc per section, so that “significant” and “improved” mean what they claim to mean:

1. **Chance-level control** (Section 6.2): random-heatmap baseline run through the identical Otsu+Dice pipeline, to separate genuine model failure from Dice’s known geometric bias toward large structures.
2. **Noise-probe diagnosis** (Sections 7.5-7.6): feeding pure synthetic noise through a trained model’s real input pipeline to check whether an apparent capability (e.g. RQ4’s improved classification accuracy) is grounded in real content or a shortcut/artifact.
3. **Cross-split replication** (Section 6.2): re-running the core finding on 2 additional independent random splits, not just reporting one lucky/unlucky split.
4. **Family-wise FDR correction** (Benjamini-Hochberg): every paired significance claim made anywhere in this report is corrected across the full accumulated family of tests, not evaluated against an uncorrected $\alpha=0.05$.
5. **Pipeline decomposition** (Section 6.3): when a metric is produced by a multi-stage pipeline, measuring each stage’s separate contribution rather than attributing the result to the stage of interest by default. Applied to the heatmap/threshold split, this showed that roughly half of the measured collapse for two of

three regions came from the binarization step rather than from grounding — a confound invisible to the chance-level control in tool 1, because the control passes through the same binarization and the artifact cancels out of the ratio.

6. **Metric triangulation** (Section 6.3): confirming a finding with a second metric whose failure modes differ from the first. Dice conflates pointing with delineation and carries a geometric size penalty; the pointing game separates them and has no such penalty. Where the two agree the claim is robust; where they diverge — as at ET-small, at chance on pointing but merely “low” on Dice — the divergence itself is the finding.
7. **Replication at the right unit** (Sections 7.8, 7.10): a within-run paired test over 213 patients answers “is this difference real *for this trained model*”, not “is this difference real”. Since every arm here is one training run, the unit of replication is the run, and the honest evidence is sign consistency across independently seeded runs measured against a retraining noise floor. Applied to RQ7 this demoted a $p \approx 6 \times 10^{-35}$ result to noise; applied to RQ5 it upgraded a reported null to a consistent positive.
8. **Metric well-definedness** (Section 7.11): checking that a metric measures what its name implies before interpreting it. The pointing game presumes a well-defined peak; a strided sliding window makes the heatmap piecewise-constant over blocks, so `argmax` returns a block corner and the “peak” is ambiguous by up to the stride. Measuring the size of the tied-maximum plateau exposed this and changed several Section 6.3 numbers substantially.

On the value of these tools. Four claims in this report were overturned or materially qualified by tools 5-8, *after* they had already been written up as findings: the magnitude of the size collapse (tool 5), the RQ7 encoder effects and the RQ5 null (tool 7), the pointing-game hit rates (tool 8), and — most consequentially — the identity of the project’s best intervention (Section 7.11, tool 5 applied to the threshold). Each was a plausible, statistically significant, internally consistent result. That is the honest lesson of this project: with a multi-stage pipeline and a shared confounded metric, significance testing alone does not protect you, because every arm inherits the same confound and the comparison still looks clean.

Figure below visualizes the outcome of every one of those corrected significance tests at once — every ablation, every region, every size bin, in one panel:

Reading this figure end to end tells the whole story at a glance: RQ3b/RQ3c (rows 3-6) are almost entirely solid blue, the isolated-window fix genuinely working almost everywhere it’s tested; RQ4 and RQ6 (rows 7-8) are almost entirely solid red, both training-side attempts genuinely underperforming the baseline despite RQ6’s improvement over RQ4; RQ5 (bottom row) is almost entirely white, exactly as a robustness check that finds no meaningful difference should look; and RQ2/RQ3 (rows 1-2) are a genuine mix, matching their more nuanced, region-dependent stories in Sections 7.1 and 7.2. See Section 7.12 for the substantive conclusion this figure supports. Note that this figure, like the leaderboard, shows Otsu-thresholded comparisons; Section 7.11 shows the RQ3b/RQ3c rows do not survive a better-calibrated binarizer.

9. Challenges

Several non-trivial obstacles came up during implementation, beyond the RQ2 negative result discussed above:

- **Kaggle API version mismatch.** A newly-generated Kaggle API access token turned out to require kaggle CLI $\geq 1.8.0$, which in turn requires Python ≥ 3.11 — incompatible with the project’s Python 3.10 environment. Resolved by using Kaggle’s legacy API key format instead, which the older, compatible CLI version supports.
- **A silent dependency collision that broke CUDA.** Installing MONAI pulled in a much newer PyTorch build (with a different CUDA version, 13.0) than the one already validated as compatible with this cluster’s GPUs (12.1). After pinning PyTorch back down, the code still failed with `libcudnn.so.9: cannot open shared object file`. The root cause: the `nvidia-*-cu12` and `nvidia-*-cu13` pip packages install their shared libraries into the *same* internal path (e.g. `nvidia/cudnn/lib/`), so uninstalling the newer package deleted the older package’s actual `.so` files while leaving its package metadata claiming it was still intact. Fixed by force-reinstalling every affected cu12 package to restore the real files, and pinning an older MONAI release that doesn’t force a PyTorch upgrade in the first place.
- **A second, unrelated version conflict.** The transformers library refuses to load PubMedBERT’s check-

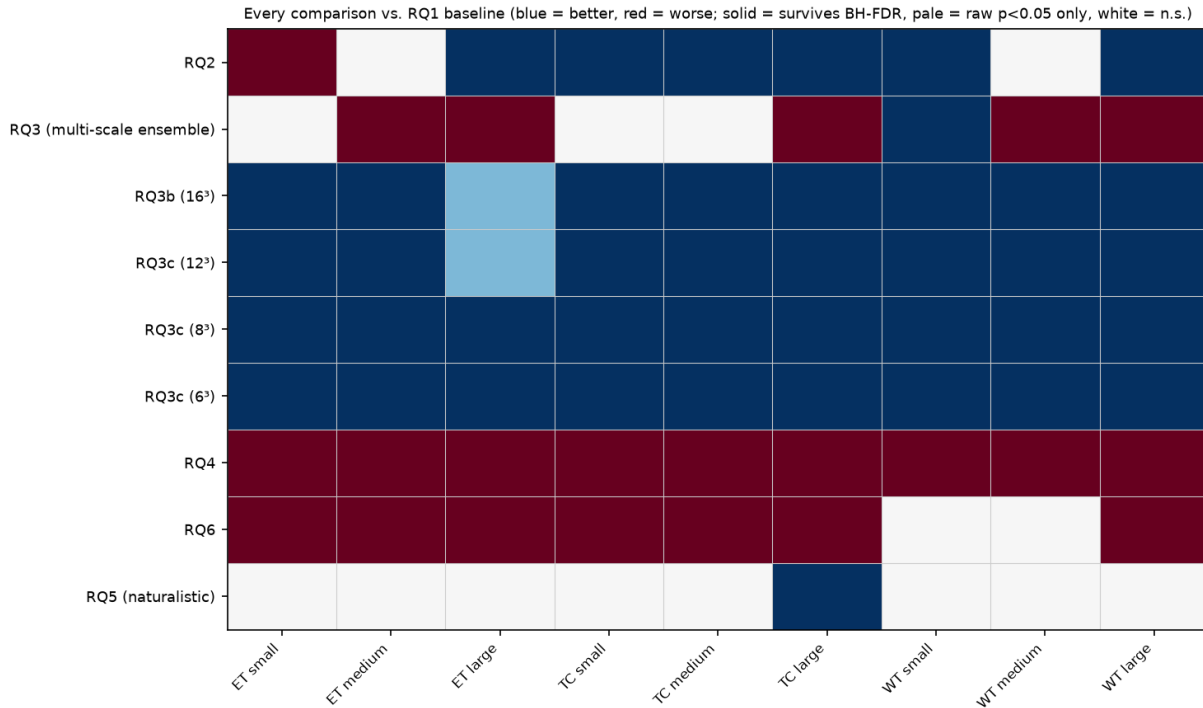


Figure 12: Every paired significance test run in this project, RQ1 baseline vs. each ablation, one row per comparison, one column per region $\times$ size bin. Blue = significantly better, red = significantly worse, white = no significant difference; solid color survives BH-FDR correction, pale color is significant only before correction.

point format (an older `pytorch_model.bin`; no `safetensors` version exists in that model’s repository) unless PyTorch ≥ 2.6 , a defensive measure against a real PyTorch CVE. Bumping PyTorch would have risked re-opening the CUDA compatibility problem above, so an older `transformers` release was pinned instead — a purely software-loading restriction, not a functional requirement.

- **A known BraTS2020 data quirk.** One patient’s segmentation file uses a non-standard filename left over from the original hospital de-identification process (Section 4), which crashed the preprocessing script partway through a 369-patient run. This also exposed a design flaw in the first version of the preprocessing script — it only wrote its summary CSV once, at the very end, so the crash would have silently discarded already-computed results for the preceding patients. Fixed by writing results incrementally per patient and adding a filename fallback for the one irregular case.
- **No GPU on the login node.** The interactive machine has no GPU at all; a CPU-only correctness test confirmed the training code was *correct* (loss decreasing, no crashes) but took over 11 minutes of CPU time for a handful of tiny batches — impractical for real training. This required learning the cluster’s SLURM job submission system (partitions, GPU resource requests, account/QOS) to actually run anything at meaningful scale, rather than treating it as optional infrastructure.
- **An architectural dead-end avoided before it was built.** Grad-CAM was the originally planned localization technique. On closer inspection, the trained model globally average-pools each patch down to a single embedding vector, meaning there is no spatial feature map left near the output for Grad-CAM to meaningfully back-propagate onto — it would have produced a near-single-voxel “heatmap.” This was caught during design, before implementation, and a sliding-window similarity map was used instead, which matches how the model actually represents space.
- **BERT anisotropy.** Raw PubMedBERT sentence embeddings for the four class descriptions turned out to have ~ 0.99 pairwise cosine similarity to each other — meaning the frozen text encoder alone carries almost

no discriminative signal, and the trainable projection head has to do essentially all the separating work during contrastive training. This was verified directly with a pairwise similarity check before committing to the training pipeline design.

10. Limitations

- **Single anatomy, single dataset.** All results are on BraTS2020 brain tumors. Whether the same size-dependent collapse and the same failure of size-conditioned prompting hold for other organs/lesion types (e.g. lung nodules, liver lesions) is untested here — Future Work below proposes this as the next check.
- **Modest per-bin sample sizes.** Size bins range from $n=20$ (WT large) to $n=32$ (WT small) patients. The overall monotonic pattern is consistent and large in effect size, but individual bin means, especially for ET (27/369 patients have no enhancing tumor at all, further shrinking that region’s usable sample), should be read with that sample size in mind.
- **Lightweight backbone, not a competitive segmentation model.** A ResNet-10 with sliding-window Otsu thresholding is a deliberately simple architecture chosen to make the size-vs-quality relationship easy to isolate and interpret. Absolute Dice numbers from the text-conditioned arms should not be compared to state-of-the-art BraTS leaderboards. Section 6.1’s P' check bounds how much of that gap is the setup rather than a pipeline defect: the same data, split and metric under conventional dense supervision reach published-range Dice, so the low text-conditioned numbers are a property of the task formulation, not of broken plumbing.
- **Templated, not naturalistic, text.** Region descriptions are hand-written templates (Section 5), not real radiologist-authored report sentences. RQ5 tested one naturalistic rewrite; across three seeds it is consistently *better* than templated text rather than equivalent (Section 7.10), though not formally significant at $n=3$ runs. Either way it is one specific naturalistic phrasing, not a sample of real report variation.
- **Replication is 3 runs, not full k-fold.** RQ1’s core pattern held in all 3 seeds (9/9 region $\times$ bin), and RQ2/RQ4/RQ5/RQ6 and all four RQ7 conditions have since been replicated across 3 seeds each (Sections 7.8, 7.10). This is a substantial strengthening over the single-run evidence, but 3 runs give the cross-seed t -tests almost no power, which is why those sections lead with sign consistency against a noise floor rather than with p -values. A full k -fold sweep would be stronger.
- **The Otsu threshold was not a neutral choice, and it changed a headline conclusion.** Otsu and the stride-16 window were chosen for transparency rather than to maximize Dice. Section 6.3 quantifies the price: $2.2\text{--}5.4\times$ in Dice relative to an oracle-volume threshold, $6\text{--}223\times$ volume over-prediction, and *anti*-correlation with true lesion size. Section 10 of an earlier draft argued the Section 7 comparisons were nonetheless internally fair because every arm shared the threshold. **That argument was wrong**, and Section 7.11 shows why: the winning intervention’s entire advantage is Otsu-specific and reverses under every better-calibrated rule. A shared confound is not a cancelled confound when arms interact with it differently.
- **The window sweep (RQ3b/RQ3c) has not been fully re-run under a better binarizer.** Section 7.11 re-scored only the 32^3 and 8^3 endpoints. The 16^3 , 12^3 and 6^3 points, and the reported ET/TC plateau at 6^3 , are still Otsu-only results and should be treated as provisional in light of that section.
- **The oracle-volume threshold is a diagnostic, not a method.** It uses the ground-truth voxel count and is therefore unavailable at inference. It is reported only to bound how much of the collapse is attributable to thresholding. The pct99 fixed-percentile rule is the deployable version, and it recovers much of the same benefit.
- **RQ6’s fix is incomplete, and its diagnostic follow-up is not a deployable method.** The uniformity regularizer fixed 2 of 3 shortcut behaviors identified by the noise probe, not all 3 — a residual hub bias remains for the smallest-scale pipeline, and RQ6 still underperforms the plain RQ1 baseline in 7 of 9 bins. Separately, the single-scale oracle test used to isolate why ensembling helps RQ6 relies on the *true* size bin label, which is not available at real inference time — it is a mechanism-isolating diagnostic, not a proposed evaluation protocol.
- **The window-size floor was only characterized down to 6^3 , and only for two of three regions.** RQ3c found a plateau for ET/TC at 6^3 , but WT was still improving significantly at that point — the true floor for WT, and whether ET/TC’s plateau holds at even smaller windows (4^3 , 2^3), is untested.

11. Future Work

- **A smarter cross-scale combination rule.** RQ3 and RQ6 together show the failure mode isn't multi-scale representation learning itself, nor is ensembling inherently harmful (RQ6 benefits from it) — but naive voxel-wise max lets the noisiest scale win when the underlying model wasn't trained to be calibrated at that scale (RQ3). Worth trying: a learned gating/attention mechanism across scales, or weighting by each scale's calibration/confidence rather than taking a raw max.
- **Finish fixing the residual hub bias from RQ6.** The uniformity regularizer fixed 2 of 3 noise-probe shortcut behaviors and improved localization over RQ4 in all 9 bins, but the smallest-scale pipeline still shows residual bias toward the "large" class, and RQ6 still trails the plain RQ1 baseline overall. Worth trying: hard-negative mining that specifically contrasts the "large" and "small" classes during training, or a stronger uniformity weight, to see if the last residual bias can be closed and RQ6 can be pushed past the RQ1 baseline rather than just past RQ4.
- **Re-run the whole window sweep under pct99.** Section 7.11 re-scored only the 32^3 and 8^3 endpoints and found the ranking inverts. The $16^3/12^3/6^3$ points and the reported ET/TC plateau are still Otsu-only and may not survive; this is now the cheapest high-value experiment left, since it needs no retraining.
- **Close the gap between P' and the text-conditioned model.** Section 6.1 shows a supervised U-Net reaches 0.64 Dice on the smallest enhancing tumors where the text-conditioned model reaches 0.01, on identical data. That $60\times$ gap is the real target, and it is now bounded rather than speculative. The obvious intermediate is a text-conditioned model with a dense decoder rather than a globally-pooled encoder, which would remove the sliding-window bottleneck entirely and make Grad-CAM-style attribution viable again.
- **Make the text pathway carry information at all.** RQ7 and RQ8 together show the query is a class index. Worth trying: contrastive negatives built from *within*-region text perturbations (true description vs. its negation as a hard negative pair), which would force the projection head to separate polarity rather than only region identity — the single cheapest change that could make the language side do work.
- **Find WT's window-size floor, and re-check ET/TC's,** at 4^3 and 2^3 — but under a calibrated threshold, given Section 7.11.
- **Extend to a second dataset** (e.g. LIDC-IDRI lung nodules) to test whether the failure pattern generalizes beyond brain tumors.

12. Effort / Contribution

This was an individual project; all design decisions, code, experiments, and writing below were done solo.

What I had to learn. Going in, I had not worked with MONAI, 3D medical image formats (NIfTI, multi-modal MRI co-registration), or SLURM job scheduling — all three were new to me and required real ramp-up: understanding how MONAI's 3D ResNet expects channel/volume layout, how to normalize and resample NIfTI volumes correctly (z-scoring within a brain mask rather than globally, which matters for skull-stripped data), and how to structure sbatch scripts against this cluster's specific partitions (dev for smoke tests capped at 10 minutes, general for real training/eval runs, correct account and GPU resource flags). I also had not previously implemented contrastive text-image (here, text-volume) alignment from scratch, so getting the shared embedding space, temperature-scaled cosine similarity loss, and sliding-window inference-time localization working correctly took real iteration — including recognizing partway through that Grad-CAM would not work at all on a globally-pooled architecture, before wasting time implementing it. On the statistics side, I had used t-tests before but not paired Wilcoxon signed-rank tests or Benjamini-Hochberg FDR correction across an accumulating family of tests, both of which turned out to be essential once the project grew past a single comparison — without FDR correction, several of the "significant" results in Section 7 would not have been defensible.

What I already knew. I came in comfortable with Python and PyTorch — writing custom Dataset/DataLoader classes, training loops, optimizers and schedulers, checkpointing, and reading loss curves to tell a bug apart from a bad hyperparameter — so none of the mechanics of building and training a model was new. I had a working understanding of transformer language models and of BERT specifically (tokenization, mean-pooling versus CLS extraction, using a frozen encoder as a feature extractor), which is why the anisotropy problem in Section 5 was something I thought to check for rather than something that ambushed me. I was already fluent with numpy and pandas for data wrangling, with matplotlib for figures, and with git. On the statistics side I was comfortable with correlation, t-tests, and the general logic of null-hypothesis testing, though not with the

non-parametric and multiple-comparison machinery this project ended up needing. I also had prior exposure to CNNs and 2D computer vision, which transferred partially — the concepts carried over, but essentially every practical detail of working in 3D (memory budgeting, patch sampling, anisotropic voxel spacing) did not.

Rough time split. Approximately: 8% reading related work (the 3D medical VLM grounding-failure literature in Section 3, plus the methods papers cited in Section 5.1); 15% environment and data setup (BraTS2020 download, preprocessing, and — unexpectedly costly — resolving the CUDA/PyTorch/transformers dependency conflicts in Section 9); 20% writing the core pipeline (text encoder, volume encoder, contrastive training loop, sliding-window localization); 12% debugging (mostly the dependency issues, plus the incremental-CSV-write bug caught before it cost a full re-run); 25% designing and running the experiment sequence (RQ1 through RQ12, including the noise probes, the P’ supervised reference, and the seed replications); 8% on the statistics layer specifically (paired testing, FDR correction across an accumulating family, bootstrap intervals, and reworking the analysis once I realized the unit of replication was the training run and not the patient); and 12% writing and revising this report.

Where the time actually went, versus where I expected. I expected the bulk of the effort to be in getting a model to work. It was not — the baseline trained on essentially the first serious attempt. The real cost was in *checking* results, and the checks were where the project’s actual content came from. Four conclusions I had already written up as findings did not survive their own follow-up test (Section 8), and the last of them — discovering that this project’s apparent best intervention won only under the threshold I had standardized on — arrived late enough that it required rewriting the report’s conclusion rather than just adding a caveat. If I ran this project again I would build the diagnostic layer first and the model second.

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